

Dissociable dynamic effects of expectation during statistical learning

Reviewed Preprint

v2 • July 21, 2025

Revised by authors

Reviewed Preprint

v1 • December 18, 2024

Hannah H McDermott , Federico De Martino, Caspar M Schwiedrzik, Ryszard Auksztulewicz

Department of Education and Psychology, Freie Universität Berlin, Berlin, Germany • Berlin School of Mind and Brain, Berlin, Germany • Faculty of Psychology and Neuroscience, Maastricht University, Maastricht, Netherlands • Neural Circuits and Cognition Lab, European Neuroscience Institute Göttingen - A Joint Initiative of the University Medical Center Göttingen and the Max Planck Institute for Multidisciplinary Sciences, Göttingen, Germany • Perception and Plasticity Group, German Primate Center, Leibniz Institute for Primate Research, Göttingen, Germany • Cognitive Neurobiology, Research Center One Health Ruhr, University Alliance Ruhr, Faculty of Biology and Biotechnology, Ruhr-University Bochum, Bochum, Germany

 https://en.wikipedia.org/wiki/Open_access
 Copyright information

eLife Assessment

This **important** study is of relevance for the fields of predictive processing, perception and learning, with a well-designed paradigm allowing the authors to avoid several common confounds in investigating predictions, such as adaptation. Using a state-of-the-art multivariate EEG approach, the authors test the opposing process theory and find evidence in support of it - i.e., the persuasive within trial effects. However, the interactions across block are not well motivated and much less persuasive, such that the support for the conclusions is only **incomplete** at present.

<https://doi.org/10.7554/eLife.103689.2.sa4>

Abstract

The brain is thought to generate internal predictions, based on previous statistical regularities in the environment, to optimise behaviour. Predictive processing has been repeatedly demonstrated and seemingly explains expectation suppression (ES), or the attenuation of neural activity in response to expected stimuli. However, the mechanisms behind ES are unclear and various models of the mechanisms supporting ES have been suggested with conflicting evidence. Sharpening models propose that expectations suppress neurons that are not tuned to the expected stimulus, increasing the signal-to-noise ratio for expected stimuli. In contrast, dampening models posit that expectations suppress neurons that are tuned to the expected stimuli, increasing the relative response amplitude for unexpected stimuli. Previous studies have used decoding analyses to examine these effects, with increases in decoding accuracy interpreted in terms of sharpening and decreases related to dampening. The opposing process theory (OPT) has suggested that both processes may occur at different time points, namely that initial sharpening is followed by later dampening of the neural representations of the expected stimulus as learning progresses. In this study we aim to test this theory and shed light on the dynamics of expectation effects, both at single trial level and over time. Thirty-one participants completed a statistical learning task consisting of paired scene categories whereby a “leading” image from one

category is quickly followed by a “trailing” image from a different category. Multivariate EEG analyses focussed on decoding stimulus information related to the trailing image category. Within-trial, decoding analyses showed that stimulus expectation increased decoding accuracy at early latencies and decreased decoding accuracy at later latencies, in line with OPT. However, across trials, stimulus expectation decreased decoding accuracy in initial trials and increased decoding accuracy in later trials. We theorise that these dissociable dynamics of expectation effects within and across trials can be explained in the context of hierarchical learning mechanisms. Our single trial results provide evidence for the OPT, while our results over time suggest that sharpening and dampening effects emerge at different stages of learning in the visual domain.

Introduction

Influential theories of perception and learning suggest a key role of prediction in both processes (1–3). However, it remains unclear whether the influence of prediction on neural activity across sources of prediction, time scales and learning stages can be explained by a single mechanism (4–7). Repetition suppression (RS), or the attenuation of neural activity in response to repeated (and therefore predictable) stimuli, has been well documented across a range of sensory modalities, species, and experimental paradigms (8,9). Predictive processing theories explain RS as a manifestation of minimising prediction errors (PEs) through adaptive changes in predictions about the content and precision of sensory inputs (8). Expectation suppression (ES), or the attenuation of neural activity in response to expected stimuli (regardless of its frequency) appears to be a related phenomenon, but ES is more controversial in terms of its underlying mechanisms. In an earlier study attempting at dissociating the two phenomena, it has been shown that RS and ES affect neural activity in separable time windows (10). More recently, however, Feuerriegel et al. (11) pointed out major challenges to the evidence underlying ES in the visual system, such as the conflation of ES with RS, surprise effects, attention effects, and stimulus novelty. While empirical evidence for ES is more fragmentary (11), it is in principle a more convincing neural signature of predictive processing, as it is less readily explained than RS by mere neural adaptation (12), which does not necessarily reflect cognitive processes.

At least two types of neural activity modulation have been linked to the effects of expectation. Sharpening models propose that expectations suppress neurons that are not tuned to the expected stimulus, thus inhibiting information inconsistent with the expectation (13,14). Depending on the level of analysis, this may result in sharper tuning curves (i.e., increased signal-to-noise ratio for expected stimuli) for a particular neuron or small population, amounting to relative expectation enhancement; and/or in sparse responses (i.e., net activity decrease) in a larger population (15). In contrast, dampening models posit that expectations suppress neurons that are tuned to the expected stimuli, effectively “explaining away” the prediction error responses (16). By cancelling information in line with expectations, uninformative activity in the sensory stream is prevented while favouring novel information (17,18). Studies attempting to settle the debate have produced mixed results on both accounts. Kok et al. (13) and Garlich & Blank (19) both found evidence for neural sharpening in the visual domain using fMRI and multivariate pattern analysis (MVPA). Another study using 7T fMRI found both suppression of irrelevant neural responses in visual areas and selective enhancement of relevant neural responses in higher-order frontoparietal cortices (20). In contrast, Richter et al. (18,21) demonstrated evidence in favour of dampening of neural responses in the visual domain. In the latter study, after training participants on a statistical learning task, stimulus features were decoded from fMRI signals separately in expected vs unexpected trials. The rationale of the analysis was that, if expectation leads to dampening, the response amplitudes will be lower, and as a result stimulus decoding should be lower in expected trials; conversely, if expectation leads to sharpening, the signal-to-noise ratio will increase, and as a result stimulus decoding should increase in expected trials. Both studies found lower decoding for expected trials, consistent with

suppressed activity of neuronal populations tuned towards expected stimuli. However, electrophysiological research in the area is scarce (17,22), despite its potential for parsing out temporal differences which may go unnoticed in fMRI studies with poor temporal resolution.

Taking temporal differences into account, the recently suggested Opposing Process Theory (OPT) has suggested that sharpening and dampening may both occur, but at different time points during the predictive process (23). This model seeks to reconcile contrasting findings as some studies show that expected events are perceived with greater intensity 50 ms after presentation but this bias reverses around 200 ms (24,25), while classically reported ES effects most often take place ~150 ms after presentation (26). OPT suggests that within each trial, initial processing relies on prior knowledge to sharpen sensory representations towards the expected stimuli, and a later processing stage follows, dampening the neural representations of the expected stimulus. This is based in part on trends in prior research. Studies have found that while expected events are perceived with greater intensity 50 ms after presentation, this bias reverses by 200 ms such that unexpected events are perceived with greater intensity (24,25). While this theory is largely untested thus far, it will prove a highly useful benchmark for future research. Namely, by implementing decoding analyses in a time-resolved manner we can better parse out the intricacies of this model. Sharpening and dampening models have previously been investigated using time-resolved methods, albeit with different study designs. Han et al. (27) found that stimulus-specific event-related activity is dampened for expected tones in the auditory cortex, while studies examining the inferior temporal (IT) cortex in monkeys have found that IT neurons represent less accurately predictable compared with unpredictable stimuli (16,17). It is worth noting that the above studies used repeated stimuli (although they did change the pairings of stimuli between trials), and trained their tasks prior to collecting data.

Theoretical work has proposed that, similar to the effects of prediction within a trial, learning should also be linked to neural activity suppression, albeit at hierarchically higher levels of neural processing (28). However, empirical work yielded more nuanced results. Previous research has focussed on the mechanisms of ES after learning has taken place, while the effects of expectation across trials (i.e., during learning) have been largely overlooked. In an earlier fMRI study (29), it has been shown that repetition effects are qualitatively different for the initial vs. later stimulus repetitions, gradually enhancing vs. suppressing BOLD activity, respectively. In contrast, EEG work has shown that neural responses are more strongly modulated by initially formed repetition-based predictions than by their subsequent revisions (30). Furthermore, in an MEG study differentiating the effects of expectation and familiarity, both factors were linked to suppressed neural activity but expectation (previously seen stimuli with expected vs. unexpected structure) attenuated activity in the lateral occipital cortex, whereas familiarity (previously seen vs. unseen stimuli with no predictable structure) additionally led to temporal sharpening of evoked responses in early visual regions (31). However, the study only characterised these effects after learning, averaging across many trials. While theoretical accounts such as the reverse hierarchy theory of perceptual learning (32) and hierarchical Bayesian framework of statistical learning (33) suggest that perceptual learning effects progress from higher to lower levels of the visual system, in classical predictive coding hierarchies the time scales increase along the processing hierarchy in an ascending manner (28). Thus, it is unclear whether dampening vs. sharpening effects (previously found to occur in anterior vs. posterior regions) (20) should be identified at earlier vs. later learning stages.

Here, we assessed the effects of expectation on EEG-based stimulus decoding during statistical learning while controlling for repetition effects, which may render the expectation effects impossible to parse out given the similarity between the two phenomena. We focused on empirically testing the Opposing Process Theory on visually evoked responses. We expected to find univariate evidence of expectation suppression, by way of reduced ERP amplitude to trailing images in valid trials over occipital channels. Furthermore, in line with the OPT, we hypothesised that sharpening effects will lead to higher decoding accuracy for valid trailing images earlier

within trials, and dampening effects will lead to higher decoding for invalid images later within trials. We also explored the dynamics of expectation effects across trials. In brief, our results show support for OPT, however with different dynamics within trials than across learning stages.

Results

Participants (N=31) were exposed to a sequence of visual scenes (**Fig. 1A**) while their neural activity was recorded using EEG (see Methods for more details). The sequence was manipulated with respect to the expectation of image categories. In each trial, participants were presented with two images from nine different categories in quick succession, with five ‘Leading’ categories, and four ‘Trailing’ categories. Over 3500 categorised images were used to avoid the repetition of individual stimuli and control for RS. Category pairs and the transitional probabilities between them were determined by the transitional probability matrix depicted in **Fig. 1B**. Each participant viewed 1728 trials without prior training and each image was only presented once.

Behavioural results

After the EEG recording, a majority of participants (N=21) were asked to perform a categorization task on images drawn from the same categories. Transitional probabilities in this task were kept the same as in the EEG session, and participants performed a speeded indoor/outdoor categorization task on the ‘Trailing’ images. A repeated measures ANOVA was conducted to compare reaction times (RTs) across conditions in order to assess implicit learning, which showed significant differences between conditions ($F_{(2,40)} = 4.231, p = 0.022, \eta^2 = 0.175$). Paired t-tests were conducted to further elucidate RT differences in between conditions. Participants reacted significantly faster in valid trials (mean = 598.9 ms, SEM = 7.2 ms) than in invalid trials (mean = 623.6 ms, SEM = 7.2 ms; $t_{20} = -2.18, p = 0.041, d = 0.476$, two-tailed). Participants also reacted significantly faster in valid trials than in neutral trials (mean = 624.7 ms, SEM = 5.7 ms; $t_{20} = 2.88, p = 0.00, d = 0.629$, two-tailed). There was no significant difference in reaction times between invalid and neutral trials ($t_{20} = 0.120, p = 0.906, d = 0.026$, two-tailed). Significant differences between both valid and invalid, and valid and neutral conditions suggest that associations between pairs were learned implicitly (**Fig 1C**).

Univariate EEG analyses

In order to assess the impact of expected vs unexpected trials on visually evoked ERPs, a paired t-test was conducted on the epoched amplitude at channel Oz. Since previous research has questioned the robustness of univariate effects of ES (**11**), we conducted a target region-of-interest analysis on the channel showing peak amplitude of the visually evoked response (channel Oz; **Fig. 2B**). Unexpected trials resulted in a significantly more pronounced negative deflection 150 ms after image onset ($t_{26} = 3.89, p = 0.023$) after FWE correction. (**Fig. 2A**). This is in line with the idea that neural responses to expected stimuli are attenuated in ES. A more conservative and exploratory whole-brain analysis, correcting for all channels and time points failed to reach significance, consistent with the issues raised by Feuerriegel et al. (**11**). The group level topography of the amplitudes differences between valid and invalid trials are represented in **Fig. 2B**.

Decoding analyses

To test the hypothesis derived from the OPT that expectation should have opposing effects at earlier (sharpening) vs. later (dampening) response latencies, we used multivariate decoding analysis. Following previous work (**21, 27**), we reasoned that sharpening should lead to increased decoding accuracy for expected stimuli, while dampening should lead to increased

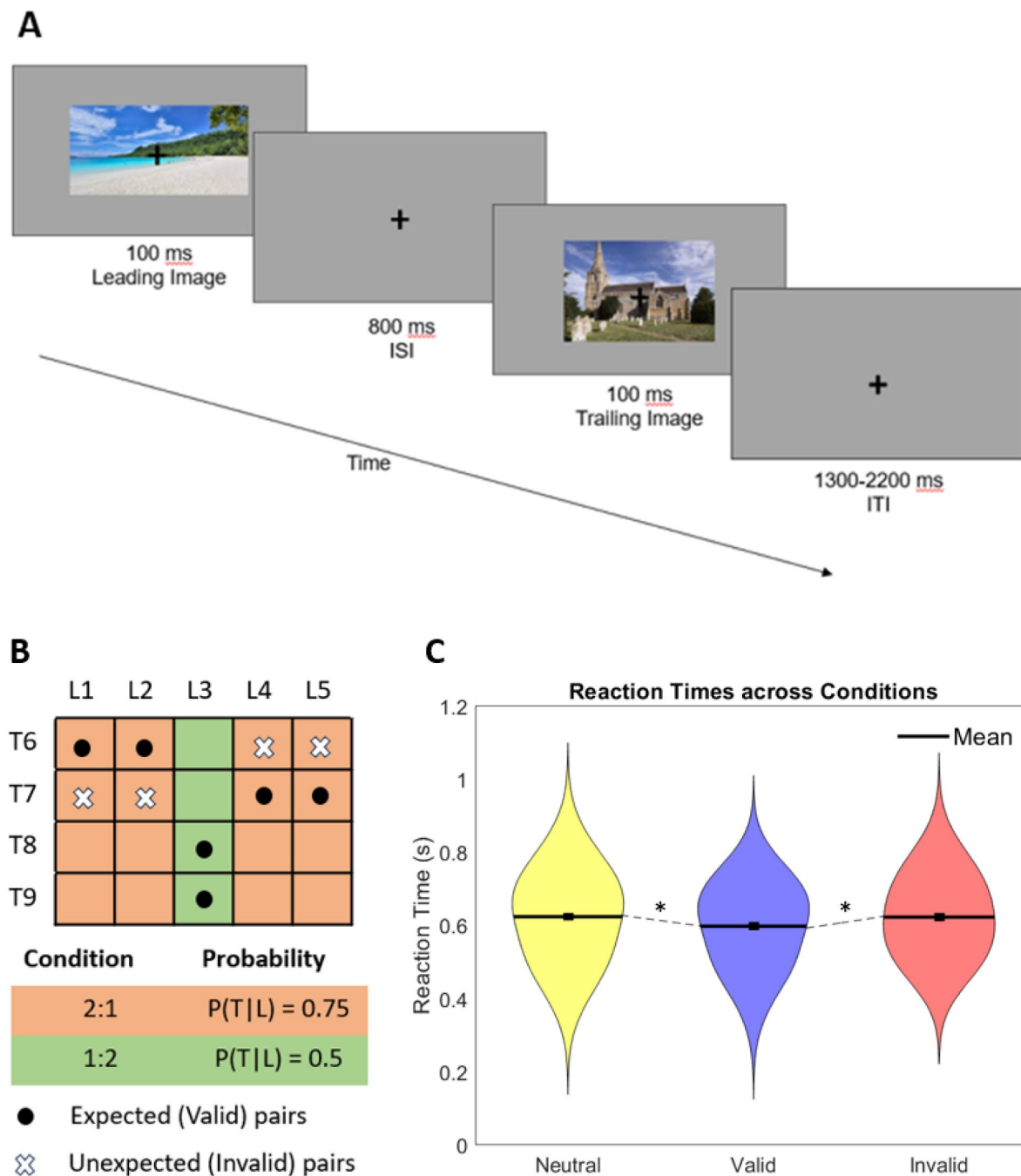


Figure 1.

Paradigm overview.

A. A single trial, with two example images. Images were presented for 100 ms each with 800 ms interstimulus interval, and an intertrial interval of 1300-2200 ms. Participants were required to respond to upside down images with button press; all upside down images were trailing. **B.** Transitional probability matrix determining category pairs. Five leading categories and four trailing categories were used. Orange represents the 2:1 condition, and green represents the 1:2 (control) condition. Cells with dots represent the valid pairs, cells with Xs represent the invalid pairs, and empty cells represent non-existent pairs. **C.** Reaction times across Valid, Invalid, and Neutral conditions.

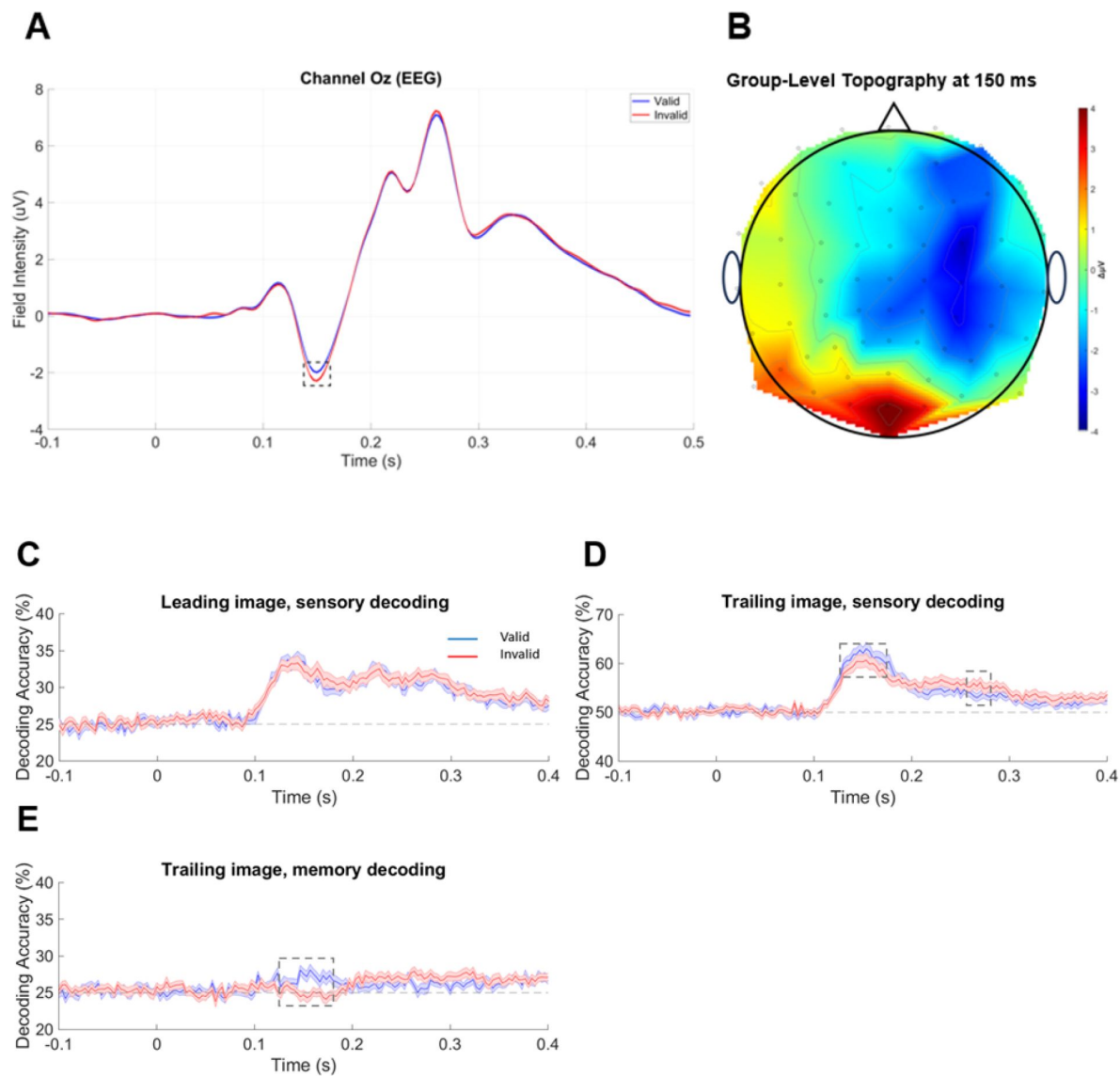


Figure 2.

Results of data analysis.

A. Grand average ERP statistical analyses at Channel Oz. Significant difference between valid and invalid trial amplitudes. **B.** Topography plot of the t-values resulting from paired t-tests comparing Valid and Invalid EEG data at 150 ms. **C.** SVM decoding results for Leading image sensory decoding. Dashed vertical line at 0 s indicates stimulus onset. **D.** Decoding results for Trailing image sensory decoding. Dashed rectangles denote latencies of significant effects (pFWE < 0.05). Dashed vertical line at 0 s indicates stimulus onset. **E.** Decoding results for Trailing image memory decoding. Dashed rectangle denotes latencies of significant effects (pFWE < 0.05). Dashed vertical line at 0 s indicates stimulus onset.

decoding accuracy for unexpected stimuli. To this end, we calculated a linear SVM-based category decoding accuracy at each time point based on EEG amplitudes, separately for valid vs. invalid trials (27 [↗](#)). Three separate decoding analyses were conducted: “sensory decoding” (decoding the visual category of the presented image, done separately for the leading and trailing images), and “memory decoding” (decoding the preceding visual category based on the trailing images). The above analyses were all subject to cluster-based family-wise error (FWE) correction for multiple comparisons across time points. FWE is a standard method of correcting for multiple comparisons and decreasing the cluster-based false positive ratio implemented in SPM. It uses random field theory assumptions (34 [↗](#)) to account for the spatiotemporal correlation between neighbouring data points. Clusters were thresholded to 0.05 uncorrected, then reported clusters which were significant at cluster-level $p_{FWE} < 0.05$.

In the sensory decoding analysis based on leading images, decoding accuracy was above chance for both valid ($T_{\max} = 4.79$, $p_{FWE} < 0.001$) and invalid trials ($T_{\max} = 4.02$, $p_{FWE} < 0.001$) from 100 ms, with no significant difference between them ($T_{\max} = 2.47$, $p_{FWE} = 0.987$) (Fig. 2C [↗](#)). This initial analysis largely served as a sanity check, since no difference between decoding the category of valid and invalid images should be observed as the leading image cannot be invalid.

On the contrary, in sensory decoding of trailing images, decoding accuracy was significantly different between valid and invalid images 123-180 ms after image onset, where decoding the category of valid images was significantly higher than invalid images (cluster-level statistics: $T_{\max} = 3.62$, $p_{FWE} < 0.001$), but decoding the category of invalid images was significantly higher than valid images 280-296 ms after image onset ($T_{\max} = 3.89$, $p_{FWE} < 0.001$) (Fig. 2D [↗](#)). Therefore, stimulus category expectation had opposing effects on trailing category decoding based on EEG signals early vs. late within a trial. The decoding accuracy of both valid ($T_{\max} = 5.10$, $p_{FWE} < 0.001$) and invalid trials ($T_{\max} = 3.83$, $p_{FWE} < 0.001$) was also above chance in this condition from 100 ms.

Finally, as with sensory decoding of trailing images, memory decoding accuracy based on valid trailing images was significantly higher than invalid trailing images 123-180 ms after image onset ($T_{\max} = 3.96$, $p_{FWE} < 0.001$) (Fig. 2E [↗](#)). Both valid ($T_{\max} = 4.67$, $p_{FWE} = 0.047$; from 100 ms) and invalid ($T_{\max} = 3.40$, $p_{FWE} < 0.001$; from 200 ms) trailing images yielded above-chance decoding.

In order to better assess how the difference in prediction and memory decoding between valid and invalid pairs changes during learning, we separated each data set into four trial bins. In order to assess the utility of examining each bin in more detail, we calculated the trial-by-trial time-series of the decoding accuracy benefit for valid vs. invalid for each participant and averaged this benefit across time points for each of the two significant time windows. Based on this, we fitted a logarithmic model to quantify the change of this benefit over trials, then found the trial index for which the change of the logarithmic fit was $< 0.1\%$, indicating stabilized decoding accuracy. Given the results of this analysis and to ensure a sufficient number of trials, we focussed our further analyses on bins 1-2 to directly assess the effects of learning.

We separately analysed the early (123-180 ms) and late (280-296 ms) within-trial time window identified in the main analysis of sensory decoding based on the trailing images (Fig. 2D [↗](#)) as well as the early (123-180 ms) time window identified in the memory decoding analysis (Fig. 2E [↗](#)). We fit each of these effects to a linear mixed effects model with Holm-Bonferroni correction for multiple comparisons across bins. Outliers larger than 3 standard deviations based on Cook's distance were removed.

In the early time window of sensory decoding, the main effects of validity ($T_9 = 0.84141$, $p = 0.403$) and bin ($T_9 = 1.9242$, $p = 0.058$) were not significant. The interaction between validity and bin was significant, however ($T_9 = -2.1679$, $p = 0.033$). Cohen's f for the full model was $f = 1.688$. Post-hoc analyses of the effects of each bin showed no significant differences between valid and invalid trials in the first bin ($T_{28} = 1.2036$, $p = 0.239$, $f = 0.052$) but significant differences in the second bin

($T_{23} = 3.8704$, $p < 0.001$, $f = 0.651$) (Fig. 3A). In other words, the decoding benefit of valid predictions - visible at early post-stimulus latencies when averaging over all trials - took approximately 15 minutes (or 50 trials per condition) to become significant.

Analyses of the later peak in sensory decoding found the opposite effect. The main effect of validity ($T_{77} = 3.5402$, $p < 0.001$), and the main effect of bin ($T_{77} = 2.2399$, $p = 0.028$) both contributed significantly. The interaction between them ($T_{77} = -2.6854$, $p = 0.009$) also showed significant differences. Cohen's f for the full model was $f = 1.5943$. In contrast to the early peak, post-hoc analyses showed that the first bin had significant differences between valid and invalid trials ($T_{24} = 4.293$, $p < 0.001$, $f = 0.768$) but the second bin did not ($T_{24} = 0.2673$, $p = 0.791$, $f = 0.003$) (Fig. 3B). In other words, the decoding benefit of invalid predictions - visible at late post-stimulus latencies when averaging over all trials - disappeared after approximately 15 minutes (or 50 trials per condition).

Finally, in the memory decoding, each of the main effects for validity ($T_{1,77} = -0.407$, $p = 0.685$), bin ($T_{1,77} = 0.751$, $p = 0.455$), and the interaction between them ($T_{1,77} = -0.794$, $p = 0.429$) failed to find significant effects after removal of outliers (Fig. 3C). Cohen's f for the full model was $f = 0.997$.

In order to examine the extent to which participants' neural responses were linked to their behavioural responses, we calculated Pearson correlation coefficients for behavioural responses (difference in RTs), neural responses (difference in decoding accuracy) during the sharpening effect, and neural responses during the dampening effect based on the difference in mean responses for each participant. Correlations between the behavioural responses and the sharpening effect ($r_{19} = 0.121$, $p = 0.602$) and between the behavioural responses and the dampening effect ($r_{19} = 0.127$, $p = 0.584$) were not significant. It should be noted however that EEG and behavioural data were not recorded simultaneously, therefore this null finding should be treated with caution.

Discussion

We set out to investigate the effects of expectation on stimulus decoding during statistical learning and demonstrated dynamic differences in decoding expected vs. unexpected stimulus categories, both within and across trials. EEG-based decoding of expected vs. unexpected visual scene categories showed that within trials, a relative decoding boost for expected stimuli was followed by a relative decoding boost for unexpected stimuli, in line with the recent Opposing Process Theory (OPT) (20). However, across trials, the decoding boost for unexpected stimuli was observed earlier than the decoding boost for expected stimuli, possibly due to a dynamic relationship between cortical processing hierarchies (32,35,36).

The OPT (23) combines both sharpening and dampening models of stimulus expectation, which have previously been mapped onto a relative decoding boost for expected vs. unexpected stimuli respectively (27). In this model, hypothesis units (units representing the brain's 'best guess' about the outside world) are initially strongly weighted towards the expected, however when agents encounter events that elicit surprise, increased gain on surprising inputs leads to high fidelity representations of unexpected events across hypothesis units. In simple terms, early sharpening (which favours processing of expected information) is followed by later dampening (which in turn allows novel information to be processed) within trials. This is supported by our within-trial results, whereby we found that decoding of expected images was significantly higher than that of unexpected images early during the trial, while the opposite effect was found later during the trial. Differences in early vs late prediction error detection have previously been observed using fMRI (5). This study used a visual detection task to demonstrate that immediate implicit (early) detection of PEs enables fast but partial comparison of bottom-up sensory input with top-down predictive information, but effortful explicit (late) processing permits more

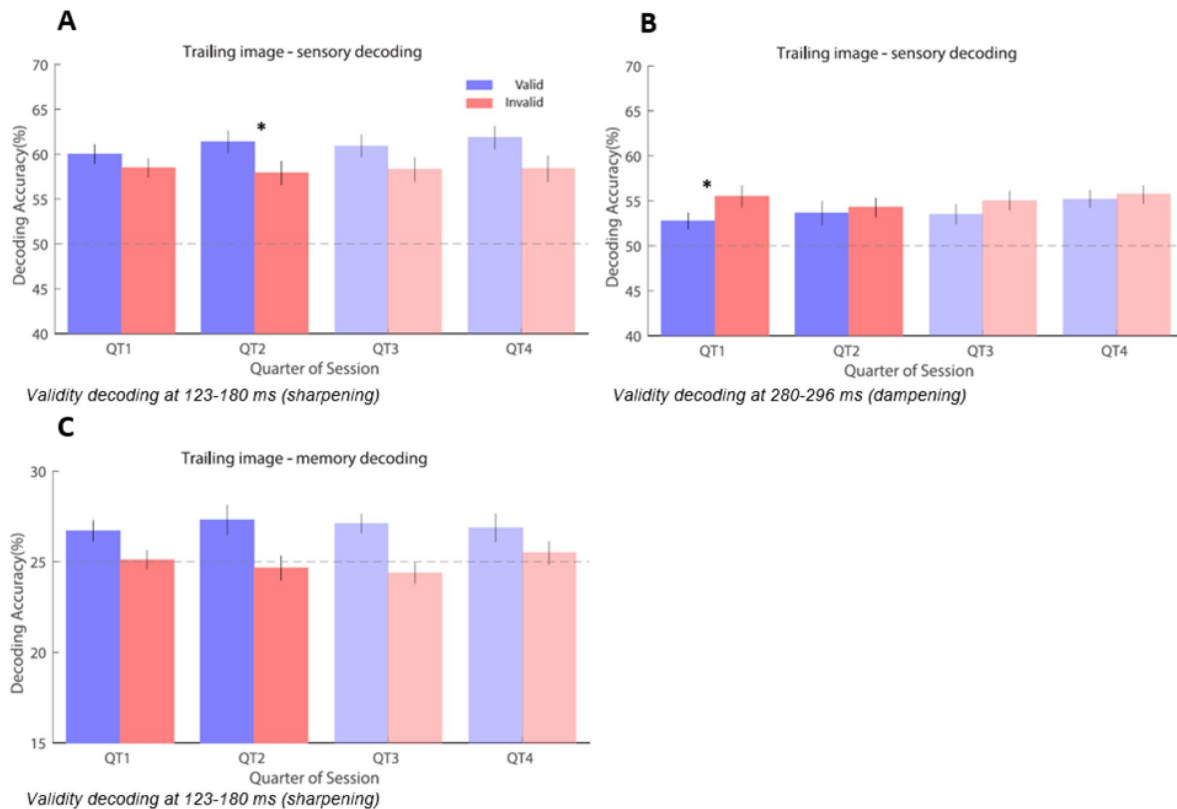


Figure 3.

Analysis of validity effects on decoding over trial bins.

Error bars indicate SEM. Dashed horizontal lines indicate chance level. **A.** Learning over time at 123-180 ms in sensory decoding. **B.** Learning over time at 280-196 ms in sensory decoding. **C.** Learning over time at 280-296 ms in memory decoding.

comprehensive assessment of the type of mismatch between expected and actual input. While the authors did not interpret their results in terms of dampening or sharpening models, it is possible that the differences observed are a result of changing temporal dynamics in the neural response which are difficult to demonstrate using fMRI. In this context, higher-level processes would modulate detection of prediction error through top-down processes and regulate adaptation of predictive models as a result.

Interestingly, the early (~150 ms) decoding accuracy boost found for expected trials coincided with the univariate ERP amplitude increase found for unexpected trials. Such seemingly opposing effects are commonly explained based on the notion that sharpening leads to fewer units responding to the expected stimulus, and the population response becoming sparser and more selective (15). According to Friston's original theoretical model assessment of predictive coding (1), sharpening and dampening occur in parallel but in separate prediction and error neurons which would reside in deep and superficial layers of the brain respectively. As such, studies that have argued for sharpening models (13,14), have focussed primarily on the response to expectations in the occipital cortex and V1, while studies that have favoured dampening models (18,21,27) have centred more on the neural response to violated expectations in the ventral visual stream. Using fMRI, Thomas et al. (36) found that expected and unexpected events are represented across the cortical column in V1. However, while expected events were represented similarly across layers, unexpected events were only well represented in superficial layers. One recent EEG study has attempted to directly elucidate the mechanisms of the OPT. Xu et al. (37) used auditory cues followed by a predictable series of flashes to investigate the mechanisms of ES. In line with OPT, they found that the N1 was augmented, while the N2 was attenuated. However, this study focussed entirely on ERP analyses (rather than multivariate decoding or gain/tuning models), which limits the interpretability of the results in terms of sharpening vs. dampening. Other studies have also found evidence for dampening after learning only. For example, two studies (16,17) examining the inferior temporal (IT) cortex in monkeys using single- and multi-unit recordings have found that IT neurons represent less accurately predictable vs. unpredictable stimuli. Similarly, MEG studies have shown that representations of expected stimuli in the neural signal appear shortly before stimulus onset (38) and that prior expectations lead to overall dampened activity in the auditory cortex (27). An fMRI study also showed overall dampened activity in the ventral visual stream (18). It is worth noting that several of the studies which showed sharpening found sharpening for predictions which were implemented as task-relevant (13,14), while some of the studies which showed dampening found effects in response to task-irrelevant predictions (18,21). However, it is unlikely that effects of task-relevance are being parsed out in the present study, where no instructions regarding relevance were given to participants, and behavioral tests were only administered after implicit learning. None of the aforementioned studies showed a reversal of sharpening and dampening effects, but it is worth noting that their experimental design did not allow for testing these effects over the course of learning. Some studies did use categorised stimuli to reduce this impact, but exemplars were still repeated multiple times over the course of the experiment (17,39), failing to fully account for the confounding effects of RS, which may mask the ES effects being examined. The present study controls for confounding RS effects, examines ES effects both during learning and after learning has occurred, and utilises the high temporal resolution of EEG, providing a unique insight into the dynamics of these phenomena.

In contrast to our within-trial results, when we examined how this process developed over multiple trials, we found that dampening occurred within the first ~15 minutes of recording and was only later followed by sharpening. We hypothesise that this is a result of a dynamic relationship between hierarchical levels of cortical processing. Previous work has shown that in MEG (40), early-latency responses typically reflect early cortical stages, while late-latency responses typically reflect higher cortical stages. Our results are in line with this in that across trials, late-latency decoding differences arise within the initial trials, suggesting that higher-order regions quickly process statistical regularities during learning. Similarly, early-latency decoding

differences occur later during the experiment, consistent with the theory that lower-order regions need more time to accumulate evidence, while the distinction between higher-order dampening and lower-order sharpening can also be observed in rapid forms of learning, as seen in rapid one-shot perceptual learning (20). In short, ‘invalid’ trials are more salient early in the experiment as priors are still being formed, which leads to predictions occurring earlier in each trial, later in the experiment. Earlier studies have pointed to a dissociation between early and late ventral stream effects, which may also fit into this framework (18,21). Other studies using statistical learning have indicated slower dynamics, in that some regions, such as the hippocampus, begin preferentially representing PEs early during learning, and predictions are preferentially represented later during learning (41). This is consistent with our finding that early dampening is followed by later sharpening over time. Another recent fMRI study found that in the presence of predictions, early stages of the processing hierarchy exhibit well separable and high-dimensional neural geometries resembling those at the top of the hierarchy, which is accompanied by a systematic shift in tuning properties from higher to lower areas, enriching lower areas with higher-order, invariant representations in place of their feedforward tuning properties (42). In the context of these studies, our results could be explained by higher-order regions quickly detecting stimulus statistics and show evidence of predictive dampening to expected inputs, and then gradually delegate those predictions to lower-order regions which start showing evidence of predictive sharpening. This suggestion is somewhat speculative however, and is thus intended to stimulate further research. It should also be noted that we demonstrate these across-trial dynamics in the visual domain only, and other domains (such as action and audition) may elicit different results and still require additional research.

In addition to the dynamic effects of expectation on sensory decoding of the trailing image, we also found that valid expectation increases decoding accuracy of the memory of the leading image. Within trials, this effect was found at similar latencies as the positive effect of expectation on sensory decoding; however, across trials, the effect on memory decoding started earlier and persisted throughout most of the experiment. A memory trace linking the trailing and leading image is a key component of associative learning (43). Furthermore, the finding that within a trial, sensory and memory decoding occur at similar latencies and are subject to similar effects of expectation, is broadly consistent with work showing simultaneous sensory, mnemonic, and predictive representations within the same neocortical regions (44,45). However, the differential dynamics of these effects across trials suggest that memory and prediction encoding is also partly dissociable, not only at the level of the hypothesised circuit mechanisms, likely involving the neocortex and the hippocampus (2), but also over time. This suggestion is consistent with a neuroimaging study showing that during associative learning, the hippocampus gradually switches from signalling PEs (i.e., mismatch between memory trace and actual stimulus) to signalling predictions (i.e., stimulus expectation independent of the actual stimulus) (41).

In summary, our study is the first to show dynamic effects of expectation on stimulus processing during learning, both within and across trials. We have shown using EEG and decoding analyses that early dampening is followed by later sharpening over time, but within trials early sharpening is followed by later dampening. This provides direct evidence for the OPT, while also indicating that sharpening and dampening effects emerge at different learning stages.

Methods

Participants

Thirty-one healthy participants (25 female, 1 non-binary, 5 male) aged 18-35 (mean = 22.7) were recruited from the student population at Freie Universität Berlin. One participant was left-handed. Participants had no history of neurological illness, were not currently taking psychotropic medication, and had normal or corrected-to-normal vision. The study was approved by the Ethics

Committee of the Department of Education and Psychology at Freie Universität Berlin. Participants were compensated with either €30, or research participation credits. Data from one participant was excluded due to a technical issue which resulted in an absence of triggers in the EEG data. The final sample size (N=30) corresponded to previous research (46 [DOI](#)).

Experimental Design and Statistical Analysis

Stimuli and experimental paradigm

Main task

Approximately 3500 colour images across nine categories were generated using artificial intelligence software Craiyon (Craiyon LLC, 2022). Images were visually inspected to ensure they appeared realistic. Some physically implausible examples were discarded. Stimuli were presented on a 38 cm LCD monitor with 1280 x 1024 resolution and 60 Hz refresh rate, the viewing distance was 62 cm, and stimuli were presented at 2.55°. The experimental paradigm was adapted from Richter et al. (18 [DOI](#)) with several modifications, as described below. Participants were exposed to pairs of visual stimuli, comprising artificially generated natural scenes drawn from different scene categories. In each trial, participants were presented with two images from different categories in quick succession, each presented for 100 ms with 800 ms interstimulus interval and 1300-2200 ms intertrial interval (**Fig. 1A** [DOI](#)). A fixation cross was presented throughout the experiment. Each image belonged to one of nine scene categories, with five 'Leading' categories ('barn', 'beach', 'library', 'restaurant', 'cave') and four 'Trailing' categories ('church', 'conference room', 'castle', 'forest'). Category pairs and the transitional probabilities between them were determined by the transitional probability matrix depicted in **Fig. 1B** [DOI](#). As previously established, many prior studies have overlooked the confounding effects of RS in their experimental paradigms. To control for RS, categorised images were used to avoid the repetition of stimuli. As such, each individual image was only presented once. In the experimental condition, categories were paired in a 2:1 manner, where two different 'Leading' categories could result in one 'Trailing' category with 75% validity. For example, both 'beach' and 'barn' as 'Leading' categories would result in 'church' as a 'Trailing' category with 75% validity. In the control condition, one 'Leading' category led to two 'Trailing' categories with 50%/50% accuracy. Participants were not informed of the associations between categories, and were instead instructed to respond by button press when images appeared upside down, which occurred in ~5% of trials. This acted as catch trials to ensure participants maintained attention on the images. The occurrence of these catch trials was randomised, as was trial order. Participants did not undergo a training session, such that the neural signatures of early learning processes may be investigated using EEG. Each participant completed eight blocks consisting of 216 trials per block, totalling ~ 90 mins of testing.

Categorisation task

EEG data was collected from the first 10 participants as a pilot version of the experiment, after which the categorisation task was added to the experiment, resulting in a lower number of participants with behavioral results. After completion of the main task, the majority of participants (n=21) performed a categorisation task to ensure learning had taken place. In this task, participants were presented with the same images as before and instructed to indicate via button press whether the 'Trailing' image takes place indoors or outdoors. This task aimed to assess implicit reaction time (RT), as the statistical regularities learned in the main task could be used to predict what category the 'Trailing' image would belong to. Participants were not informed of the intent behind this task or instructed to make use of what they had learned in the main task.

Data Analysis

Categorisation task analysis

Behavioural data from the categorisation task was analysed in terms of RTs. All RTs exceeding 2 SD above or below the mean across subjects were excluded as outliers. RTs for valid, invalid, and neutral trials underwent a log transformation before being averaged separately per participant. A repeated measures ANOVA was performed to compare RTs in each condition, then three separate paired t-tests were conducted comparing valid vs. invalid, valid vs. neutral, and invalid vs. neutral. Removal of RTs exceeding 2 SDs above or below the mean resulted in two participants having no neutral trials, thus these empty values were replaced with the mean of the remaining subjects. In order to examine the extent to which participants' neural responses were linked to their behavioural responses, we calculated Pearson correlation coefficients for behavioural responses, neural responses during the sharpening effect (123-180 ms), and neural responses during the dampening effect (280-296 ms) based on the difference in mean decoding across time within both time windows for each participant.

EEG data preprocessing

EEG data were acquired at 2048 Hz from 64 electrodes using a BioSemi ActiveTwo system. Electrodes were arranged according to the international 10/20 system. Preprocessing was conducted using custom SPM12 (Wellcome Trust Centre for Neuroimaging, University College London; RRID: SCR_007037) for Matlab (The MathWorks; RRID: SCR_001622) as well as custom Matlab scripts. Continuous EEG data were high-pass filtered above 0.1 Hz and notch-filtered between 48-52 Hz using 5th-order zero-phase Butterworth filters. Data were then downsampled to 300 Hz. Blinks were detected using two horizontal and two vertical electro-oculogram (EOG) electrodes placed above and below the left eye, and to the left and right of the eyes. An epoch of -200 ms to 400 ms relative to artefact onset was applied to detected eye blinks and spatiotemporal confounds were calculated using Singular Value Decomposition (SVD). Top two principal components were removed from the segments of data associated with each eye blink (47). Data were then further denoised using the Dynamic Separation of Sources (48) which maximises the reproducibility of stimulus-evoked responses across trials in order to increase the signal-to-noise ratio of ERPs. During this process, data were re-referenced offline to the average of all channels. Continuous data were epoched between 200 ms before leading stimulus onset to 2600 ms after stimulus onset. Each epoch was baseline-corrected to the mean of the pre-stimulus onset (from -200 ms to 0 ms relative to stimulus onset). Epochs with an average root mean square (RMS) amplitude exceeding the median by two standard deviations (SDs) were excluded from analysis in order to remove contamination by transient artefacts. This resulted in rejecting 11.2% ($\pm 1.03\%$ the standard error of the mean (SEM) across participants) trials on average. Data were then averaged across trials and a low-pass filter of 48 Hz was applied. Catch trials were removed prior to analysis.

Decoding analyses

Single-trial data underwent decoding based on linear support vector machines (SVM) implemented using custom MATLAB code. As part of feature selection, Principal Component Analysis (PCA) was conducted to reduce dimensionality of sensor-level EEG data, and subsequent components were selected based on signal-to-noise (SNR) with a cutoff threshold of 8 dB. As a result, an average of 7.6 components, SD = 0.498, were used across participants. Given the poor spatial resolution of EEG, considering all channels for feature extraction increases the computational cost, risk of overfitting, and required number of trials. As such, PCA can be useful for dimensionality reduction prior to SVM-based decoding (49,50). Three separate decoding analyses were conducted, decoding the visual category of both leading and trailing images, decoding the predictable trailing images based on the leading images, and memory decoding for trailing images. Valid and invalid were analysed separately at each step. In both analyses (of valid vs invalid trials), the decoder was trained and tested using leave-8-out cross-validation, such that

in each test set, single trials corresponding to all 8 unique combinations of 4 leading and 2 trailing images were included (see **Fig. 1B** [↗](#)). Visual category decoding was used to differentiate neural responses to leading vs. trailing categories, while memory decoding aimed to assess memory of the leading category based on the trailing category.

The initial analysis aimed at quantifying decoding as a function of peristimulus time. To this end, we calculated the decoding accuracy per time point, using all trials. The resulting decoding time series were converted to one-dimensional images, entered into a GLM, outliers larger than 3 standard deviations based on Cook's distance were removed, and subjected to (1 [↗](#)) a one-sample t-test against chance-level, equivalent to a fixed-effects analysis (51 [↗](#)), and (2 [↗](#)) a paired t-test between valid and invalid categories. Given that EEG time-series are autocorrelated over time, statistical tests were corrected for multiple comparisons over time using cluster-based FWE correction. A follow-up analysis aimed at quantifying changes in decoding over learning. To this end, based on the time windows in which we found significant decoding differences between valid and invalid trials (see below), we averaged the decoding accuracy estimates within each time window, separately for four bins of trials. In order to assess the utility of examining each bin in more detail, we calculated the trial-by-trial time-series of the decoding accuracy benefit for valid vs. invalid for each participant and averaged this benefit across time points for each of the two significant time windows, resulting in 200 trials per participant. Based on this, we fitted a logarithmic model to quantify the change of this benefit over trials, then found the trial index for which the change of the logarithmic fit was $< 0.1\%$. In the early time window (123-180 ms) the median proportion of trials after which the decoding benefit stabilises is 11%, with the 99th percentile being reached after 38% of trials. In the later time window (280-196 ms), the median proportion of trials after which the decoding benefit stabilises is 18%, with the 99th percentile being reached after 47% of trials. Given that decoding stabilises after approximately 50% of trials, and in order to balance this with having a sufficient number of trials for analysis, we focussed our further analyses on bins 1-2 to directly assess the effects of learning. Each bin encompassed approximately 15 minutes of testing or 432 trials on average.

Data and Code Availability

Datasets and scripts used are available at https://github.com/hannahmcderm/Expectation_EEG_eLife [↗](#).

Additional information

Funding

German Research Foundation Individual Research Grant (AU 423/2-1)

References

1. Friston K (2005) **A theory of cortical responses** *Philos Trans R Soc B Biol Sci* **360**:815–36 [Google Scholar](#)
2. Jiang LP, Rao RPN (2024) **Dynamic predictive coding: A model of hierarchical sequence learning and prediction in the neocortex** *PLOS Comput Biol* **20**:e1011801 [Google Scholar](#)
3. Walsh KS, McGovern DP, Clark A, O’Connell RG (2020) **Evaluating the neurophysiological evidence for predictive processing as a model of perception** *Ann N Y Acad Sci* **1464**:242–68 [Google Scholar](#)
4. Blom T, Bode S, Hogendoorn H (2021) **The time-course of prediction formation and revision in human visual motion processing** *Cortex* **138**:191–202 [Google Scholar](#)
5. Malekshahi R, Seth A, Papanikolaou A, Mathews Z, Birbaumer N, Verschure PFMJ, et al. (2016) **Differential neural mechanisms for early and late prediction error detection** *Sci Rep* **6**:24350 [Google Scholar](#)
6. Wang R, Shen Y, Tino P, Welchman AE, Kourtzi Z (2017) **Learning Predictive Statistics: Strategies and Brain Mechanisms** *J Neurosci* **37**:8412–27 [Google Scholar](#)
7. Wischniewski M, Schutter DJLG (2019) **Electrophysiological correlates of prediction formation in anticipation of reward- and punishment-related feedback signals** *Psychophysiology* **56**:e13379 [Google Scholar](#)
8. Aukstulewicz R, Friston K (2016) **Repetition suppression and its contextual determinants in predictive coding** *Cortex* **80**:125–40 [Google Scholar](#)
9. Grotheer M, Kovács G (2016) **Can predictive coding explain repetition suppression?** *Cortex* **80**:113–24 [Google Scholar](#)
10. Todorovic A, de Lange FP (2012) **Repetition suppression and expectation suppression are dissociable in time in early auditory evoked fields** *J Neurosci* **32**:13389–95 [Google Scholar](#)
11. Feuerriegel D, Vogels R, Kovács G (2021) **Evaluating the evidence for expectation suppression in the visual system** *Neurosci Biobehav Rev* **126**:368–81 [Google Scholar](#)
12. Barron HC, Garvert MM, Behrens TEJ (2016) **Repetition suppression: a means to index neural representations using BOLD?** *Philos Trans R Soc B Biol Sci* **371**:20150355 [Google Scholar](#)
13. Kok P, Jehee JFM, de Lange FP (2012) **Less is more: Expectation sharpens representations in the primary visual cortex** *Neuron* **75**:265–70 [Google Scholar](#)
14. Yon D, Gilbert SJ, de Lange FP, Press C. (2018) **Action sharpens sensory representations of expected outcomes** *Nat Commun.* **9**:4288 [Google Scholar](#)
15. Kremláček J, Kreegipuu K, Tales A, Astikainen P, Pöldver N, Näätänen R, et al. (2016) **Visual mismatch negativity (vMMN): A review and meta-analysis of studies in psychiatric and**

neurological disorders *Cortex* **80**:76–112 [Google Scholar](#)

16. Kumar S, Kaposvari P, Vogels R (2017) **Encoding of predictable and unpredictable stimuli by inferior temporal cortical neurons** *J Cogn Neurosci* **29**:1445–54 [Google Scholar](#)
17. Meyer T, Olson CR (2011) **Statistical learning of visual transitions in monkey inferotemporal cortex** *Proc Natl Acad Sci* **108**:19401–6 [Google Scholar](#)
18. Richter D, Ekman M, Lange FP de (2018) **Suppressed sensory response to predictable object stimuli throughout the ventral visual stream** *J Neurosci* **38**:7452–61 [Google Scholar](#)
19. Garlichs A, Blank H (2024) **Prediction error processing and sharpening of expected information across the face-processing hierarchy** *Nat Commun* **15**:3407 [Google Scholar](#)
20. González-García C, He BJ (2021) **A Gradient of Sharpening Effects by Perceptual Prior across the Human Cortical Hierarchy** *J Neurosci* **41**:167–78 [Google Scholar](#)
21. Richter D, Heilbron M, de Lange FP (2022) **Dampened sensory representations for expected input across the ventral visual stream** *Oxf Open Neurosci* **1**:kvac013 [Google Scholar](#)
22. Schwiedrzik CM, Freiwald WA (2017) **High-Level Prediction Signals in a Low-Level Area of the Macaque Face-Processing Hierarchy** *Neuron* **96**:89–97 [Google Scholar](#)
23. Press C, Kok P, Yon D. (2020) **The Perceptual Prediction Paradox** *Trends Cogn Sci.* **24**:13–24 [Google Scholar](#)
24. Press C, Gherri E, Heyes C, Eimer M. (2010) **Action Preparation Helps and Hinders Perception of Action** *J Cogn Neurosci.* **22**:2198–211 [Google Scholar](#)
25. Yon D, Press C. (2017) **Predicted action consequences are perceptually facilitated before cancellation** *J Exp Psychol Hum Percept Perform* **43**:1073–83 [Google Scholar](#)
26. Hughes G, Desantis A, Waszak F (2013) **Mechanisms of intentional binding and sensory attenuation: the role of temporal prediction, temporal control, identity prediction, and motor prediction** *Psychol Bull* **139**:133–51 [Google Scholar](#)
27. Han B, Mostert P, de Lange FP (2019) **Predictable tones elicit stimulus-specific suppression of evoked activity in auditory cortex** *NeuroImage* **200**:242–9 [Google Scholar](#)
28. Kiebel SJ, Daunizeau J, Friston KJ (2008) **A Hierarchy of Time-Scales and the Brain** *PLOS Comput Biol* **4**:e1000209 [Google Scholar](#)
29. Müller NG, Strumpf H, Scholz M, Baier B, Melloni L (2013) **Repetition Suppression versus Enhancement—It’s Quantity That Matters** *Cereb Cortex* **23**:315–22 [Google Scholar](#)
30. Fitzgerald K, Aukstulewicz R, Provost A, Paton B, Howard Z, Todd J (2021) **Hierarchical Learning of Statistical Regularities over Multiple Timescales of Sound Sequence Processing: A Dynamic Causal Modeling Study** *J Cogn Neurosci* **33**:1549–62 [Google Scholar](#)
31. Manahova ME, Mostert P, Kok P, Schoffelen JM, de Lange FP (2018) **Stimulus Familiarity and Expectation Jointly Modulate Neural Activity in the Visual Ventral Stream** *J Cogn Neurosci* **30**:1366–77 [Google Scholar](#)

32. Ahissar M, Hochstein S (2004) **The reverse hierarchy theory of visual perceptual learning** *Trends Cogn Sci* **8**:457–64 [Google Scholar](#)
33. Fiser J, Lengyel G (2019) **A common probabilistic framework for perceptual and statistical learning** *Curr Opin Neurobiol* **58**:218–28 [Google Scholar](#)
34. Kilner JM, Kiebel SJ, Friston KJ (2005) **Applications of random field theory to electrophysiology** *Neurosci Lett* **374**:174–8 [Google Scholar](#)
35. Nigam T, Schwiedrzik CM (2024) **Predictions enable top-down pattern separation in the macaque face-processing hierarchy** *Nat Commun* **15** [Google Scholar](#)
36. Thomas ER, Haarsma J, Nicholson J, Yon D, Kok P, Press C. (2024) **Predictions and errors are distinctly represented across V1 layers** *Curr Biol.* **4**:p2265–2271 [Google Scholar](#)
37. Xu M, Meng J, Yu H, Jung TP, Ming D (2020) **Dynamic Brain Responses Modulated by Precise Timing Prediction in an Opposing Process** *Neurosci Bull* **37**:70–80 [Google Scholar](#)
38. Kok P, Mostert P, de Lange FP (2017) **Prior expectations induce prestimulus sensory templates** *Proc Natl Acad Sci* **114**:10473–8 [Google Scholar](#)
39. Sherman BE, Graves KN, Huberdeau DM, Quraishi IH, Damisah EC, Turk-Browne NB (2022) **Temporal dynamics of competition between statistical learning and episodic memory in intracranial recordings of human visual cortex** *J Neurosci* **42**:9053–68 [Google Scholar](#)
40. Cichy RM, Pantazis D, Oliva A (2014) **Resolving human object recognition in space and time** *Nat Neurosci* **17**:455–62 [Google Scholar](#)
41. Aitken F, Kok P (2022) **Hippocampal representations switch from errors to predictions during acquisition of predictive associations** *Nat Commun* **13**:3294 [Google Scholar](#)
42. Riels K, Ramos Campagnoli R, Thigpen N, Keil A (2022) **Oscillatory brain activity links experience to expectancy during associative learning** *Psychophysiology* **59**:e13946 [Google Scholar](#)
43. Libby A, Buschman TJ (2021) **Rotational dynamics reduce interference between sensory and memory representations** *Nat Neurosci* **24**:715–26 [Google Scholar](#)
44. Cappotto D, Kang H, Li K, Melloni L, Schnupp J, Auksztulewicz R (2022) **Simultaneous mnemonic and predictive representations in the auditory cortex** *Curr Biol* **32**:2548–2555 [Google Scholar](#)
45. Barron HC, Auksztulewicz R, Friston K (2020) **Prediction and memory: A predictive coding account** *Prog Neurobiol* **192** [Google Scholar](#)
46. Huang YT, Wu CT, Fang YXM, Fu CK, Koike S, Chao ZC (2024) **Crossmodal hierarchical predictive coding for audiovisual sequences in the human brain** *Commun Biol* **7** [Google Scholar](#)
47. Ille N, Berg P, Scherg M (2002) **Artifact Correction of the Ongoing EEG Using Spatial Filters Based on Artifact and Brain Signal Topographies** *J Clin Neurophysiol* **19**:113 [Google Scholar](#)
48. de Cheveigné A, Simon JZ (2008) **Denoising based on spatial filtering** *J Neurosci Methods* **171**:331–9 [Google Scholar](#)

49. Zhang T, Chen W, Li M (2019) **Classification of inter-ictal and ictal EEGs using multi-basis MODWPT, dimensionality reduction algorithms and LS-SVM: A comparative study** *Biomed Signal Process Control* **47**:240–51 [Google Scholar](#)
50. Rahman MA, Hossain MF, Hossain M, Ahmmed R. (2020) **Employing PCA and t-statistical approach for feature extraction and classification of emotion from multichannel EEG signal** *Egypt Inform J.* **21**:23–35 [Google Scholar](#)
51. Allefeld C, Görden K, Haynes JD (2016) **Valid population inference for information-based imaging: From the second-level t-test to prevalence inference** *NeuroImage* **141**:378–392 [Google Scholar](#)

Author information

Hannah H McDermott

Department of Education and Psychology, Freie Universität Berlin, Berlin, Germany, Berlin School of Mind and Brain, Berlin, Germany, Faculty of Psychology and Neuroscience, Maastricht University, Maastricht, Netherlands
ORCID iD: [0009-0005-5399-2917](#)

For correspondence: h.mcdermott@fu-berlin.de

Federico De Martino

Faculty of Psychology and Neuroscience, Maastricht University, Maastricht, Netherlands
ORCID iD: [0000-0002-0352-0648](#)

Caspar M Schwiedrzik

Neural Circuits and Cognition Lab, European Neuroscience Institute Göttingen - A Joint Initiative of the University Medical Center Göttingen and the Max Planck Institute for Multidisciplinary Sciences, Göttingen, Germany, Perception and Plasticity Group, German Primate Center, Leibniz Institute for Primate Research, Göttingen, Germany, Cognitive Neurobiology, Research Center One Health Ruhr, University Alliance Ruhr, Faculty of Biology and Biotechnology, Ruhr-University Bochum, Bochum, Germany
ORCID iD: [0000-0003-0661-8859](#)

Ryszard Auksztulewicz

Department of Education and Psychology, Freie Universität Berlin, Berlin, Germany, Faculty of Psychology and Neuroscience, Maastricht University, Maastricht, Netherlands
ORCID iD: [0000-0001-9078-3667](#)

Editors

Reviewing Editor

Clare Press

University College London, London, United Kingdom

Senior Editor

Huan Luo

Peking University, Beijing, China

Reviewer #1 (Public review):**Summary:**

In this lovely paper, McDermott and colleagues tackle an enduring puzzle in the cognitive neuroscience of perceptual prediction. Though many scientists agree that top-down predictions shape perception, previous studies have yielded incompatible results - with studies showing 'sharpened' representations of expected signals, and others showing a 'dampening' of predictable signals to relatively enhance surprising prediction errors. To deepen the paradox further, it seems like there are good reasons that we would want to see both influences on perception in different contexts.

Here, the authors aim to test one possible resolution to this 'paradox' - the opposing process theory (OPT). This theory makes distinct predictions about how the timecourse of 'sharpening' and 'dampening' effects should unfold. The researchers present a clever twist on a leading-trailing perceptual prediction paradigm, using AI to generate a large dataset of test and training stimuli, so that it is possible to form expectations about certain categories without repeating any particular stimuli. This provides a powerful way of distinguishing expectation effects from repetition effects - a perennial problem in this line of work.

Using EEG decoding, the researchers find evidence to support the OPT. Namely, they find that neural encoding of expected events is superior in earlier time ranges (sharpening-like) followed by a relative advantage for unexpected events in later time ranges (dampening-like). On top of this, the authors also show that these two separate influences may emerge differently in different phases of learning - with superior decoding of surprising prediction errors being found more in early phases of the task, and enhanced decoding of predicted events being found in the later phases of the experiment.

Strengths:

As noted above, a major strength of this work lies in important experimental design choices. Alongside removing any possible influence of repetition suppression mechanisms in this task, the experiment also allows us to see how effects emerge in 'real time' as agents learn to make predictions. This contrasts with many other studies in this area - where researchers 'over-train' expectations into observers to create the strongest possible effects, or rely on prior knowledge that was likely to be crystallised outside the lab.

Weaknesses:

This study reveals a great deal about how certain neural representations are altered by expectation and learning on shorter and longer timescales, so I am loath to describe certain limitations as 'weaknesses'. But one limitation inherent in this experimental design is that, by focusing on implicit, task-irrelevant predictions, there is not much opportunity to connect the predictive influences seen at the neural level to perceptual performance itself (e.g., how participants make perceptual decisions about expected or unexpected events, or how these events are detected or appear).

<https://doi.org/10.7554/eLife.103689.2.sa3>

Reviewer #2 (Public review):**Summary:**

There are two accounts in the literature that propose that expectations suppress activity of neurons that are (a) not tuned to the expected stimulus to increase the signal-to-noise ratio for expected stimuli (sharpening model) or (b) tuned to the expected stimulus to highlight

novel information (dampening model). One recent account, the opposing process theory, brings the two models together and suggests that both processes occur, but at different time points: initial sharpening is followed by later dampening of the neural activity of the expected stimulus. In this study, the authors aim to test the opposing process theory in a statistical learning task by applying multivariate EEG analyses and find evidence for the opposing process theory based on the within-trial dynamics.

Strengths:

This study addresses a very timely research question about the underlying mechanisms of expectation suppression. The applied EEG decoding approach offers an elegant way to investigate the temporal characteristics of expectation effects. A strength of the study lies in the experimental design that aims to control for repetition effects, one of the common confounds in prediction suppression studies. The reported results are novel in the field and have the potential to improve our understanding of expectation suppression in visual perception.

Weaknesses:

Although some of the findings are in line with the opposing process theory, especially the EEG results only partly support the hypothesis. While the initial dampening effect occurs in the grand average ERP and in image memory decoding, the expected later sharpening effect is lacking. Moreover, some methodological decisions still remain arbitrary. One of the interesting aspects of the study - prediction decoding - had to be removed due to the fact that it could not be disentangled from category decoding. This weakens the overall scope and impact of the manuscript.

<https://doi.org/10.7554/eLife.103689.2.sa2>

Reviewer #3 (Public review):

Summary:

In their study McDermott et al. investigate the neurocomputational mechanism underlying sensory prediction errors. They contrast two accounts: representational sharpening and dampening. Representational sharpening suggests that predictions increase the fidelity of the neural representations of expected inputs, while representational dampening suggests the opposite (decreased fidelity for expected stimuli). The authors performed decoding analyses on EEG data, showing that first expected stimuli could be better decoded (sharpening), followed by a reversal during later response windows where unexpected inputs could be better decoded (dampening). These results are interpreted in the context of opposing process theory (OPT), which suggests that such a reversal would support perception to be both veridical (i.e., initial sharpening to increase the accuracy of perception) and informative (i.e., later dampening to highlight surprising, but informative inputs).

Strengths:

The topic of the present study is of significant relevance for the field of predictive processing. The experimental paradigm used by McDermott et al. is well designed, allowing the authors to avoid several common confounds in investigating predictions, such as stimulus familiarity and adaptation. The introduction of the manuscript provides a well written summary of the main arguments for the two accounts of interest (sharpening and dampening), as well as OPT. Overall, the manuscript serves as a good overview of the current state of the field.

Weaknesses:

In my opinion some details of the methods, results and manuscript raise some doubts about the reliability of the reported findings. Key concerns are:

(1) In the previous round of comments, I noted that: "I am not fully convinced that Figures 3A/B and the associated results support the idea that early learning stages result in dampening and later stages in sharpening. The inference made requires, in my opinion, not only a significant effect in one-time bin and the absence of an effect in other bins. Instead to reliably make this inference one would need a contrast showing a difference in decoding accuracy between bins, or ideally an analysis not contingent on seemingly arbitrary binning of data, but a decrease (or increase) in the slope of the decoding accuracy across trials. Moreover, the decoding analyses seem to be at the edge of SNR, hence making any interpretation that depends on the absence of an effect in some bins yet more problematic and implausible". The authors responded: "we fitted a logarithmic model to quantify the change of the decoding benefit over trials, then found the trial index for which the change of the logarithmic fit was $< 0.1\%$. Given the results of this analysis and to ensure a sufficient number of trials, we focused our further analyses on bins 1-2". However, I do not see how this new analysis addresses the concern that the conclusion highlights differences in decoding performance between bins 1 and 2, yet no contrast between these bins are performed. While I appreciate the addition of the new model, in my current understanding it does not solve the problem I raised. I still believe that if the authors wish to conclude that an effect differs between two bins they must contrast these directly and/or use a different appropriate analysis approach.

Relatedly, the logarithmic model fitting and how it justifies the focus on analysis bin 1-2 needs to be explained better, especially the rationale of the analysis, the choice of parameters (e.g., why logarithmic, why change of logarithmic fit $< 0.1\%$ as criterion, etc), and why certain inferences follow from this analysis. Also, the reporting of the associated results seems rather sparse in the current iteration of the manuscript.

(2) A critical point the authors raise is that they investigate the buildup of expectations during training. They go on to show that the dampening effect disappears quickly, concluding: "the decoding benefit of invalid predictions [...] disappeared after approximately 15 minutes (or 50 trials per condition)". Maybe the authors can correct me, but my best understanding is as follows: Each bin has 50 trials per condition. The 2:1 condition has 4 leading images, this would mean ~ 12 trials per leading stimulus, 25% of which are unexpected, so ~ 9 expected trials per pair. Bin 1 represents the first time the participants see the associations. Therefore, the conclusion is that participants learn the associations so rapidly that ~ 9 expected trials per pair suffice to not only learn the expectations (in a probabilistic context) but learn them sufficiently well such that they result in a significant decoding difference in that same bin. If so, this would seem surprisingly fast, given that participants learn by means of incidental statistical learning (i.e. they were not informed about the statistical regularities). I acknowledge that we do not know how quickly the dampening/sharpening effects develop, however surprising results should be accompanied with a critical evaluation and exceptionally strong evidence (see point 1). Consider for example the following alternative account to explain these results. Category pairs were fixed across and within participants, i.e. the same leading image categories always predicted the same trailing image categories for all participants. Some category pairings will necessarily result in a larger representational overlap (i.e., visual similarity, etc.) and hence differences in decoding accuracy due to adaptation and related effects. For example, house $\square$ barn will result in a different decoding performance compared to coffee cup $\square$ barn, simply due to the larger visual and semantic similarity between house and barn compared to coffee cup and barn. These effects should occur upon first stimulus presentation, independent of statistical learning, and may attenuate over time e.g., due to increasing familiarity with the categories (i.e., an overall attenuation leading to smaller between condition differences) or pairs.

(3) In response to my previous comment, why the authors think their study may have found different results compared to multiple previous studies (e.g. Han et al., 2019; Kumar et al., 2017; Meyer and Olson, 2011), particularly the sharpening to dampening switch, the authors emphasize the use of non-repeated stimuli (no repetition suppression and no familiarity confound) in their design. However, I fail to see how familiarity or RS could account for the absence of sharpening/dampening inversion in previous studies.

First, if the authors argument is about stimulus novelty and familiarity as described by Feuerriegel et al., 2021, I believe this point does not apply to the cited studies. Feuerriegel et al., 2021 note: "Relative stimulus novelty can be an important confound in situations where expected stimulus identities are presented often within an experiment, but neutral or surprising stimuli are presented only rarely", which indeed is a critical confound. However, none of the studies (Han et al., 2019; Richter et al., 2018; Kumar et al., 2017; Meyer and Olson, 2011) contained this confound, because all stimuli served as expected and unexpected stimuli, with the expectation status solely determined by the preceding cue. Thus, participants were equally familiar with the images across expectation conditions.

Second, for a similar reason the authors argument for RS accounting for the different results does not hold either in my opinion. Again, as Feuerriegel et al. 2021 correctly point out: "Adaptation-related effects can mimic ES when the expected stimuli are a repetition of the last-seen stimulus or have been encountered more recently than stimuli in neutral expectation conditions." However, it is critical to consider the precise design of previous studies. Taking again the example of Han et al., 2019; Kumar et al., 2017; Meyer and Olson, 2011. To my knowledge none of these studies contained manipulations that would result in a more frequent or recent repetition of any specific stimulus in the expected compared to unexpected condition. The crucial manipulation in all these previous studies is not that a single stimulus or stimulus feature (which could be subject to familiarity or RS) determines the expectation status, but rather the transitional probability (i.e. cue-stimulus pairing) of a particular stimulus given the cue. Therefore, unless I am missing something critical, simple RS seems unlikely to differ between expectation condition in the previous studies and hence seems implausible to account for differences in results compared to the current study.

Moreover, studies cited by the authors (e.g. Todorovic & de Lange, 2012) showed that RS and ES are separable in time, again making me wonder how avoiding stimulus repetition should account for the difference in the present study compared to previous ones. I am happy to be corrected in my understanding, but with the currently provided arguments by the authors I do not see how RS and familiarity can account for the discrepancy in results.

I agree with the authors that stimulus familiarity is a clear difference compared to previous designs, but without a valid explanation why this should affect results I find this account rather unsatisfying. I see the key difference in that the authors manipulated category predictability, instead of exemplar prediction - i.e. searching for a car instead of your car. However, if results in support of OPT would indeed depend on using novel images (i.e. without stimulus repetition), would this not severely limit the scope of the account and hence also its relevance? Certainly, the account provided by the authors casts the net wider and tries to explain visual prediction. Relatedly, if OPT only applies during training, as the authors seem to argue, would this again not significantly narrow the scope of the theory? Combined these two caveats would seem to demote the account from a general account of prediction and perception to one about perception during very specific circumstances. In my understanding the appeal of OPT is that it accounts for multiple challenges faced by the perceptual system, elegantly integrating them into a cohesive framework. Most of this would be lost by claiming that OPT's primary prediction would only apply to specific circumstances - novel stimuli during learning of predictions. Moreover, in the original formulation of the account, as outlined by Press et al., I do not see any particular reason why it should be limited to these specific circumstances. This does of course not mean that the present results are

incorrect, however it does require an adequate discussion and acknowledgement in the manuscript.

Impact:

McDermott et al. present an interesting study with potentially impactful results. However, given my concerns raised in this and the previous round of comments, I am not entirely convinced of the reliability of the results. Moreover, the difficulty of reconciling some of the present results with previous studies highlights the need for more convincing explanations of these discrepancies and a stronger discussion of the present results in the context of the literature.

<https://doi.org/10.7554/eLife.103689.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public reviews:

Reviewer 1 (Public Review):

Many thanks for the positive and constructive feedback on the manuscript.

This study reveals a great deal about how certain neural representations are altered by expectation and learning on shorter and longer timescales, so I am loath to describe certain limitations as 'weaknesses'. But one limitation inherent in this experimental design is that, by focusing on implicit, task-irrelevant predictions, there is not much opportunity to connect the predictive influences seen at the neural level to the perceptual performance itself (e.g., how participants make perceptual decisions about expected or unexpected events, or how these events are detected or appear).

Thank you for the interesting comment. We now discuss the limitation of task-irrelevant prediction. In brief, some studies which showed sharpening found that task demands were relevant, while some studies which showed dampening were based on task-irrelevant predictions, but it is unlikely that task relevance - which was not manipulated in the current study - would explain the switch between sharpening and dampening that we observe within and across trials.

The behavioural data that is displayed (from a post-recording behavioural session) shows that these predictions do influence perceptual choice - leading to faster reaction times when expectations are valid. In broad strokes, we may think that such a result is broadly consistent with a 'sharpening' view of perceptual prediction, and the fact that sharpening effects are found in the study to be larger at the end of the task than at the beginning. But it strikes me that the strongest test of the relevance of these (very interesting) EEG findings would be some evidence that the neural effects relate to behavioural influences (e.g., are participants actually more behaviourally sensitive to invalid signals in earlier phases of the experiment, given that this is where the neural effects show the most 'dampening' a.k.a., prediction error advantage?)

Thank you for the suggestion. We calculated Pearson's correlation coefficients for behavioural responses (difference in mean reaction times), neural responses during the sharpening effect (difference in decoding accuracy), and neural responses during the dampening effect for each participant, which resulted in null findings.

Reviewer 2 (Public Review):

Thank you for your helpful and constructive comments on the manuscript.

The strength in controlling for repetition effects by introducing a neutral (50% expectation) condition also adds a weakness to the current version of the manuscript, as this neutral condition is not integrated into the behavioral (reaction times) and EEG (ERP and decoding) analyses. This procedure remained unclear to me. The reported results would be strengthened by showing differences between the neutral and expected (valid) conditions on the behavioral and neural levels. This would also provide a more rigorous check that participants had implicitly learned the associations between the picture category pairings.

Following the reviewer's suggestion, we have included the neutral condition in the behavioural analysis and performed a repeated measures ANOVA on all three conditions.

It is not entirely clear to me what is actually decoded in the prediction condition and why the authors did not perform decoding over trial bins in prediction decoding as potential differences across time could be hidden by averaging the data. The manuscript would generally benefit from a more detailed description of the analysis rationale and methods.

In the original version of the manuscript, prediction decoding aimed at testing if the upcoming stimulus category can be decoded from the response to the preceding (leading) stimulus. However, in response to the other Reviewers' comments we have decided to remove the prediction decoding analysis from the revised manuscript as it is now apparent that prediction decoding cannot be separated from category decoding based on pixel information.

Finally, the scope of this study should be limited to expectation suppression in visual perception, as the generalization of these results to other sensory modalities or to the action domain remains open for future research.

We have clarified the scope of the study in the revised manuscript.

Reviewer 3 (Public Review):

Thank you for the thought-provoking and interesting comments and suggestions.

(1) The results in Figure 2C seem to show that the leading image itself can only be decoded with ~33% accuracy (25% chance; i.e. ~8% above chance decoding). In contrast, Figure 2E suggests the prediction (surprisingly, valid or invalid) during the leading image presentation can be decoded with ~62% accuracy (50% chance; i.e. ~12% above chance decoding). Unless I am misinterpreting the analyses, it seems implausible to me that a prediction, but not actually shown image, can be better decoded using EEG than an image that is presented on-screen.

Following this and the remaining comments by the Reviewer (see below), we have decided to remove the prediction analysis from the manuscript. Specifically, we have focused on the Reviewer's concern that it is implausible that image prediction would be better decoded than an image that is presented on-screen. This led us to perform a control analysis, in which we tried to decode the leading image category based on pixel values alone (rather than on EEG responses). Since this decoding was above chance, we could not rule out the possibility that

EEG responses to leading images reflect physical differences between image categories. This issue does not extend to trailing images, as the results of the decoding analysis based on trailing images are based on accuracy comparisons between valid and invalid trials, and thus image features are counterbalanced. We would like to thank the Reviewer for raising this issue

(2) The "prediction decoding" analysis is described by the authors as "decoding the predictable trailing images based on the leading images". How this was done is however unclear to me. For each leading image decoding the predictable trailing images should be equivalent to decoding validity (as there were only 2 possible trailing image categories: 1 valid, 1 invalid). How is it then possible that the analysis is performed separately for valid and invalid trials? If the authors simply decode which leading image category was shown, but combine L1+L2 and L4+L5 into one class respectively, the resulting decoder would in my opinion not decode prediction, but instead dissociate the representation of L1+L2 from L4+L5, which may also explain why the time-course of the prediction peaks during the leading image stimulus-response, which is rather different compared to previous studies decoding predictions (e.g. Kok et al. 2017). Instead for the prediction analysis to be informative about the prediction, the decoder ought to decode the representation of the trailing image during the leading image and inter-stimulus interval. Therefore I am at present not convinced that the utilized analysis approach is informative about predictions.

In this analysis, we attempted to decode (from the response to leading images) which trailing categories ought to be presented. The analysis was split between trials where the expected category was indeed presented (valid) vs. those in which it was not (invalid). The separation of valid vs invalid trials in the prediction decoding analysis served as a sanity check as no information about trial validity was yet available to participants. However, as mentioned above, we have decided to remove the "prediction decoding" analysis based on leading images as we cannot disentangle prediction decoding from category decoding.

(3) I may be misunderstanding the reported statistics or analyses, but it seems unlikely that >10 of the reported contrasts have the exact same statistic of $T_{max} = 2.76$. Similarly, it seems implausible, based on visual inspection of Figure 2, that the T_{max} for the invalid condition decoding (reported as $T_{max} = 14.903$) is substantially larger than for the valid condition decoding (reported as $T_{max} = 2.76$), even though the valid condition appears to have superior peak decoding performance. Combined these details may raise concerns about the reliability of the reported statistics.

Thank you for bringing this to our attention. This copy error has now been rectified.

(4) The reported analyses and results do not seem to support the conclusion of early learning resulting in dampening and later stages in sharpening. Specifically, the authors appear to base this conclusion on the absence of a decoding effect in some time-bins, while in my opinion a contrast between time-bins, showing a difference in decoding accuracy, is required. Or better yet, a non-zero slope of decoding accuracy over time should be shown (not contingent on post-hoc and seemingly arbitrary binning).

Thank you for the helpful suggestion. We have performed an additional analysis to address this issue, we calculated the trial-by-trial time-series of the decoding accuracy benefit for valid vs. invalid for each participant and averaged this benefit across time points for each of the two significant time windows. Based on this, we fitted a logarithmic model to quantify the change of this benefit over trials, then found the trial index for which the change of the logarithmic fit was < 0.1% (i.e., accuracy was stabilized). Given the results of this analysis and to ensure a sufficient number of trials, we focussed our further analyses on bins 1-2 to

directly assess the effects of learning. This is explained in more detail in the revised manuscript .

(5) The present results both within and across trials are difficult to reconcile with previous studies using MEG (Kok et al., 2017; Han et al., 2019), single-unit and multi-unit recordings (Kumar et al., 2017; Meyer & Olson 2011), as well as fMRI (Richter et al., 2018), which investigated similar questions but yielded different results; i.e., no reversal within or across trials, as well as dampening effects with after more training. The authors do not provide a convincing explanation as to why their results should differ from previous studies, arguably further compounding doubts about the present results raised by the methods and results concerns noted above.

The discussion of these findings has been expanded in the revised manuscript . In short, the experimental design of the above studies did not allow for an assessment of these effects prior to learning. Several of them also used repeated stimuli (albeit some studies changed the pairings of stimuli between trials), potentially allowing for RS to confound their results.

Recommendations for the Authors:

Reviewer 1 (Recommendations for the authors):

(1) On a first read, I was initially very confused by the statement on p.7 that each stimulus was only presented once - as I couldn't then work out how expectations were supposed to be learned! It became clear after reading the Methods that expectations are formed at the level of stimulus category (so categories are repeated multiple times even if exemplars are not). I suspect other readers could have a similar confusion, so it would be helpful if the description of the task in the 'Results' section (e.g., around p.7) was more explicit about the way that expectations were generated, and the (very large) stimulus set that examples are being drawn from.

Following your suggestion, we have clarified the paradigm by adding details about the categories and the manner in which expectations are formed.

(2) p.23: the authors write that their 1D decoding images were "subjected to statistical inference amounting to a paired t-test between valid and invalid categories". What is meant by 'amounting to' here? Was it a paired t-test or something statistically equivalent? If so, I would just say 'subjected to a paired t-test' to avoid any confusion, or explaining explicitly which statistic inference was done over.

We have rephrased this as “subjected to (1) a one-sample t-test against chance-level, equivalent to a fixed-effects analysis, and (2) a paired t-test”.

Relatedly, this description of an analysis amounting to a 'paired t-test' only seems relevant for the sensory decoding and memory decoding analyses (where there are validity effects) rather than the prediction decoding analysis. As far as I can tell the important thing is that the expected image category can be decoded, not that it can be decoded better or worse on valid or invalid trials.

In the previous version of the manuscript, the comparison of prediction decoding between valid and invalid trials was meant as a sanity check. However, in response to the other Reviewers' comments we have decided to remove the prediction decoding analysis from the revised manuscript due to confounds.

It would be helpful if authors could say a bit more about how the statistical inferences were done for the prediction decoding analyses and the 'condition against baseline' contrasts (e.g.,

when it is stated that decoding accuracy in valid trials **,in general,** is above 0 at some cluster-wise corrected value). My guess is that this amounts to something like a one-sample t-test - but it may be worth noting that one-sample t-tests on information measures like decoding accuracy cannot support population-level inference, because these measures cannot meaningfully be below 0 (see Allefeld et al, 2016).

When testing for decoding accuracy against baseline, we used one-sample t-tests against chance level (rather than against 0) throughout the manuscript. We now clarify in the manuscript that this corresponds to a fixed-effects analysis (Allefeld et al., 2016). In contrast, when testing for differences in decoding accuracy between valid and invalid conditions, we used paired-sample t-tests. As mentioned above, the prediction decoding analysis has been removed from the analysis.

(3) By design, the researchers focus on implicit predictive learning which means the expectations being formed are (by definition) task-irrelevant. I thought it could be interesting if the authors might speculate in the discussion on how they think their results may or may not differ when predictions are deployed in task-relevant scenarios - particularly given that some studies have found sharpening effects do not seem to depend on task demands (e.g., Kok et al, 2012 ; Yon et al, 2018) while other studies have found that some dampening effects do seem to depend on what the observer is attending to (e.g., Richter et al, 2018) . Do these results hint at a possible explanation for why this might be? Even if the authors think they don't, it might be helpful to say so!

Thank you for the interesting comment. We have expanded on this in the revised manuscript.

Reviewer 2 (Recommendations for the authors):

Methods/results

(1) The goal of this study is the assessment of expectation effects during statistical learning while controlling for repetition effects, one of the common confounds in prediction suppression studies (see, Feuerriegel et al., 2021). I agree that this is an important aspect and I assume that this was the reason why the authors introduced the P=0.5 neutral condition (Figure 1B, L3). However, I completely missed the analyses of this condition in the manuscript. In the figure caption of Figure 1C, it is stated that the reaction times of the valid, invalid, and neutral conditions are shown, but only data from the valid and invalid conditions are depicted. To ensure that participants had built up expectations and had learned the pairing, one would not only expect a difference between the valid and invalid conditions but also between the valid and neutral conditions. Moreover, it would also be important to integrate the neutral condition in the multivariate EEG analysis to actually control for repetition effects. Instead, the authors constructed another control condition based on the arbitrary pairings. But why was the neutral condition not compared to the valid and invalid prediction decoding results? Besides this, I also suggest calculating the ERP for the neutral condition and adding it to Figure 2A to provide a more complete picture.

As mentioned above, we have included the neutral condition in the behavioural analysis, as outlined in the revised manuscript. We have also included a repeated measures ANOVA on all 3 conditions. The purpose of the neutral condition was not to avoid RS, but rather to provide a control condition. We avoided repetition by using individual, categorised stimuli. Figure 1C has been amended to include the neutral condition). In response to the remaining comments, we have decided to remove the prediction decoding analysis from the manuscript.

(2) One of the main results that is taken as evidence for the OPT is that there is higher decoding accuracy for valid trials (indicate sharpening) early in the trial and higher

decoding accuracy for invalid trials (indicate dampening) later in the trial. I would have expected this result for prediction decoding that surprisingly showed none of the two effects. Instead, the result pattern occurred in sensory decoding only, and partly (early sharpening) in memory decoding. How do the authors explain these results? Additionally, I would have expected similar results in the ERP; however, only the early effect was observed. I missed a more thorough discussion of this rather complex result pattern. The lack of the opposing effect in prediction decoding limits the overall conclusion that needs to be revised accordingly.

Since sharpening vs. dampening rests on the comparison between valid and invalid trials, evidence for sharpening vs. dampening could only be obtained from decoding based on responses to trailing images. In prediction decoding (removed from the current version), information about the validity of the trial is not yet available. Thus, our original plan was to compare this analysis with the effects of validity on the decoding of trailing images (i.e. we expected valid trials to be decoded more accurately after the trailing image than before). The results of the memory decoding did mirror the sensory decoding of the trailing image in that we found significantly higher decoding accuracy of the valid trials from 123-180 ms. As with the sensory decoding, there was a tendency towards a later flip (280-296 ms) where decoding accuracy of invalid trials became nominally higher, but this effect did not reach statistical significance in the memory decoding.

(3) To increase the comprehensibility of the result pattern, it would be helpful for the reader to clearly state the hypotheses for the ERP and multivariate EEG analyses. What did you expect for the separate decoding analyses? How should the results of different decoding analyses differ and why? Which result pattern would (partly, or not) support the OPT?

Our hypotheses are now stated in the revised manuscript.

(4) I was wondering why the authors did not test for changes during learning for prediction decoding. Despite the fact that there were no significant differences between valid and invalid conditions within-trial, differences could still emerge when the data set is separated into bins. Please test and report the results.

As mentioned above, we have decided to remove the prediction decoding analysis from the current version of the manuscript.

(5) To assess the effect of learning the authors write: 'Given the apparent consistency of bins 2-4, we focused our analyses on bins 1-2.' Please explain what you mean by 'apparent consistency'. Did you test for consistency or is it based on descriptive results? Why do the authors not provide the complete picture and perform the analyses for all bins? This would allow for a better assessment of changes over time between valid and invalid conditions. In Figure 3, were valid and invalid trials different in any of the QT3 or QT4 bins in sensory or memory encoding?

We have performed an additional analysis to address this issue. The reasoning behind the decision to focus on bins 1-2 is now explained in the revised manuscript. In short, fitting a learning curve to trial-by-trial decoding estimates indicates that decoding stabilizes within <50% of the trials. To quantify changes in decoding occurring within these <50% of the trials while ensuring a sufficient number of trials for statistical comparisons, we decided to focus on bins 1-2 only.

(6) Please provide the effect size for all statistical tests.

Effect sizes have now been provided.

(7) Please provide exact p-values for non-significant results and significant results larger than 0.001.

Exact p-values have now been provided.

(8) Decoding analyses: I suppose there is a copy/paste error in the T-values as nearly all T-values on pages 11 and 12 are identical (2.76) leading to highly significant p-values (0.001) as well as non-significant effects (>0.05). Please check.

Thank you for bringing this to our attention. This error has now been corrected.

(9) Page 12: There were some misleading phrases in the result section. To give one example: 'control analyses was slightly above change' - this sounds like a close to non-significant effect, but it was indeed a highly significant effect of $p < 0.001$. Please revise.

This phrase was part of the prediction decoding analysis and has therefore been removed.

(10) Sample size: How was the sample size of the study be determined (N=31)? Why did only a subgroup of participants perform the behavioral categorization task after the EEG recording? With a larger sample, it would have been interesting to test if participants who showed better learning (larger difference in reaction times between valid and invalid conditions) also showed higher decoding accuracies.

This has been clarified in the revised manuscript. In short, the larger sample size of N=31 was based on previous research; ten participants were initially tested as part of a pilot which was then expanded to include the categorisation task.

(11) I assume catch trials were removed before data analyses?

We have clarified that catch trials were indeed removed prior to analyses.

(12) Page 23, 1st line: 'In each, the decoder...' Something is missing here.

Thank you for bringing this to our attention, this sentence has now been rephrased as "In both valid and invalid analyses" in the revised manuscript.

Discussion

(1) The analysis over multiple trials showed dampening within the first 15 min followed by sharpening. I found the discussion of this finding very lengthy and speculative (page 17). I recommend shortening this part and providing only the main arguments that could stimulate future research.

Thank you for the suggestion. Since Reviewer 3 has requested additional details in this part of the discussion, we have opted to keep this paragraph in the manuscript. However, we have also made it clearer that this section is relatively speculative and the arguments provided for the across trials dynamics are meant to stimulate further research.

(2) As this task is purely perceptual, the results support the OPT for the area of visual perception. For action, different results have been reported. Suppression within-trial has been shown to be larger for expected than unexpected features of action targets and suppression even starts before the start of the movement without showing any evidence

for sharpening (e.g., Fuehrer et al., 2022, PNAS). For suppression across trials, it has been found that suppression decreases over the course of learning to associate a sensory consequence to a specific action (e.g., Kiltner et al., 2019, eLife). Therefore, expectation suppression might function differently in perception and action (an area that still requires further research). Please clarify the scope of your study and results on perceptual expectations in the introduction, discussion, and abstract.

We have clarified the scope of the study in the revised manuscript.

Figures

(1) Figure 1A: Add 't' to the arrow to indicate time.

This has been rectified.

(2) Figure 3: In the figure caption, sensory and memory decoding seem to be mixed up. Please correct. Please add what the dashed horizontal line indicates.

Thank you for bringing this to our attention, this has been rectified.

Reviewer 3 (Recommendations for the authors):

I applaud the authors for a well-written introduction and an excellent summary of a complicated topic, giving fair treatment to the different accounts proposed in the literature. However, I believe a few additional studies should be cited in the Introduction, particularly time-resolved studies such as Han et al., 2019; Kumar et al., 2017; Meyer and Olson, 2011. This would provide the reader with a broader picture of the current state of the literature, as well as point the reader to critical time-resolved studies that did not find evidence in support of OPT, which are important to consider in the interpretation of the present results.

The introduction has been expanded to include the aforementioned studies in the revised manuscript.

Given previous neuroimaging studies investigating the present phenomenon, including with time-resolved measures (e.g. Kok et al., 2017; Han et al., 2019; Kumar et al., 2017; Meyer & Olson 2011), why do the authors think that their data, design, or analysis allowed them to find support for OPT but not previous studies? I do not see obvious modifications to the paradigm, data quantity or quality, or the analyses that would suggest a superior ability to test OPT predictions compared to previous studies. Given concerns regarding the data analyses (see points below), I think it is essential to convincingly answer this question to convince the reader to trust the present results.

The most obvious alteration to the paradigm is the use of non-repeated stimuli. Each of the above time-resolved studies utilised repeated stimuli (either repeated, identical stimuli, or paired stimuli where pairings are changed but the pool of stimuli remains the same), allowing for RS to act as a confound as exemplars are still presented multiple times. By removing this confound, it is entirely plausible that we may find different time-resolved results given that it has been shown that RS and ES are separable in time (Todorovic & de Lange, 2012). We also test during learning rather than training participants on the task beforehand. By foregoing a training session, we are better equipped to assess OPT predictions as they emerge. In our across-trial results, learning appears to take place after approximately 15 minutes or 432 trials, at which point dampening reverses to sharpening. Had we trained the participants prior to testing, this effect would have been lost.

What is actually decoded in the "prediction decoding" analysis? The authors state that it is "decoding the predictable trailing images based on the leading images" (p.11). The associated chance level (Figure 2E) is indicated as 50%. This suggests that the classes separated by the SVM are T6 vs T7. How this was done is however unclear. For each leading image decoding the predictable trailing images should be equivalent to decoding validity (as there are only 2 possible trailing images, where one is the valid and the other the invalid image). How is it then possible that the analysis is performed separately for valid and invalid trials? Are the authors simply decoding which leading image was shown, but combine L1+L2 and L4+L5 into one class respectively? If so, this needs to be better explained in the manuscript. Moreover, the resulting decoder would in my opinion not decode the predicted image, but instead learn to dissociate the representation of L1+L2 from L4+L5, which may also explain why the time course of the prediction peaks during the leading image stimulus-response, which is rather different compared to previous studies decoding (prestimulus) predictions (e.g. Kok et al. 2017). If this is indeed the case, I find it doubtful that this analysis relates to prediction. Instead for the prediction analysis to be informative about the predicted image the authors should, in my opinion, train the decoder on the representation of trailing images and test it during the prestimulus interval.

As mentioned above, the prediction decoding analysis has been removed from the manuscript. The prediction decoding analysis was intended as a sanity check, as validity information was not yet available to participants.

Related to the point above, were the leading/trailing image categories and their mapping to L1, L2, etc. in Figure 1B fixed across subjects? I.e. "'beach' and 'barn' as 'Leading' categories would result in 'church' as a 'Trailing' category with 75% validity" (p.20) for all participants? If so, this poses additional problems for the interpretation of the analysis discussed in the point above, as it may invalidate the control analyses depicted in Figure 2E, as systematic differences and similarities in the leading image categories could account for the observed results.

Image categories and their mapping were indeed fixed across participants. While this may result in physical differences and similarities between images influencing results, counterbalancing categories across participants would not have addressed this issue. For example, had we swapped "beach" with "barn" in another participant, physical differences between images may still be reflected in the prediction decoding. On the other hand, counterbalancing categories across trials was not possible given our aim of examining the initial stages of learning over trials. Had we changed the mappings of categories throughout the experiment for each participant, we would have introduced reversal learning and nullified our ability to examine the initial stages of learning under flat priors. In any case, the prediction decoding analysis has been removed from the manuscript, as outlined above.

Why was the neutral condition L3 not used for prediction decoding? After all, if during prediction decoding both the valid and invalid image can be decoded, as suggested by the authors, we would also expect significant decoding of T8/T9 during the L3 presentation.

In the neutral condition, L3 was followed by T8 vs. T9 with 50% probability, precluding prediction decoding. While this could have served as an additional control analysis for EEG-based decoding, we have opted for removing prediction decoding from the analysis. However, in response to the other Reviewers' comments, the neutral condition has now been included in the behavioral analysis.

The following concern may arise due to a misunderstanding of the analyses, but I found the results in Figures 2C and 2E concerning. If my interpretation is correct, then these results suggest that the leading image itself can only be decoded with ~33% accuracy (25% chance; i.e. ~8% above chance decoding). In contrast, the predicted (valid or invalid) image during the leading image presentation can be decoded with ~62% accuracy (50% chance; i.e. ~12% above chance decoding). Does this seem reasonable? Unless I am misinterpreting the analyses, it seems implausible to me that a prediction but not actually shown image can be better decoded than an on-screen image. Moreover, to my knowledge studies reporting decoding of predictions can (1) decode expectations just above chance level (e.g. Kok et al., 2017; which is expected given the nature of what is decoded) and (2) report these prestimulus effects shortly before the anticipated stimulus onset, and not coinciding with the leading image onset ~800ms before the predicted stimulus onset. For the above reasons, the key results reported in the present manuscript seem implausible to me and may suggest the possibility of problems in the training or interpretation of the decoding analysis. If I misunderstood the analyses, the analysis text needs to be refined. If I understood the analyses correctly, at the very least the authors would need to provide strong support and arguments to convince the reader that the effects are reliable (ruling out bias and explaining why predictions can be decoded better than on-screen stimuli) and sensible (in the context of previous studies showing different time-courses and results).

As explained above, we have addressed this concern by performing an additional analysis, implementing decoding based on image pixel values. Indeed we could not rule out the possibility that “prediction” decoding reflected stimulus differences between leading images.

Relatedly, the authors use the prestimulus interval (-200 ms to 0 ms before predicted stimulus onset) as the baseline period. Given that this period coincides with prestimulus expectation effects (Kok et al., 2017), would this not result in a bias during trailing image decoding? In other words, the baseline period would contain an anticipatory representation of the expected stimulus (Kok et al., 2017), which is then subtracted from the subsequent EEG signal, thereby allowing the decoder to pick up on this “negative representation” of the expected image. It seems to me that a cleaner contrast would be to use the 200ms before leading image onset as the baseline.

The analysis of trailing images aimed at testing specific hypotheses related to differences between decoding accuracy in valid vs. invalid trials. Since the baseline was by definition the same for both kinds of trials (since information about validity only appears at the onset of the trailing image), changing the baseline would not affect the results of the analysis. Valid and invalid trials would have the same prestimulus effect induced by the leading image.

Again, maybe I misunderstood the analyses, but what exactly are the statistics reported on p. 11 onward? Why is the reported T_{max} identical for multiple conditions, including the difference between conditions? Without further information this seems highly unlikely, further casting doubts on the rigor of the applied methods/analyses. For example: “In the sensory decoding analysis based on leading images, decoding accuracy was above chance for both valid (T_{max}= 2.76, pFWE < 0.001) and invalid trials (T_{max}= 2.76, pFWE < 0.001) from 100 ms, with no significant difference between them (T_{max}= 2.76, pFWE > 0.05) (Fig. 2C)” (p.11).

Thank you for bringing this to our attention. As previously mentioned, this copy error has been rectified in the revised manuscript.

Relatedly, the statistics reported below in the same paragraph also seem unusual. Specifically, the Tmax difference between valid and invalid conditions seems unexpectedly large given visual inspection of the associated figure: "The decoding accuracy of both valid (Tmax = 2.76, pFWE < 0.001) and invalid trials (Tmax = 14.903, pFWE < 0.001)" (p.12). In fact, visual inspection suggests that the largest difference should probably be observed for the valid not invalid trials (i.e. larger Tmax).

This copy error has also been rectified in the revised manuscript.

Moreover, multiple subsequent sections of the Results continue to report the exact same Tmax value. I will not list all appearances of "Tmax = 2.76" here but would recommend the authors carefully check the reported statistics and analysis code, as it seems highly unlikely that >10 contrasts have exactly the same Tmax. Alternatively, if I misunderstand the applied methods, it would be essential to better explain the utilized method to avoid similar confusion in prospective readers.

This error has also now been rectified. As mentioned above the prediction decoding analysis has been removed.

I am not fully convinced that Figures 3A/B and the associated results support the idea that early learning stages result in dampening and later stages in sharpening. The inference made requires, in my opinion, not only a significant effect in one-time bin and the absence of an effect in other bins. Instead to reliably make this inference one would need a contrast showing a difference in decoding accuracy between bins, or ideally an analysis not contingent on seemingly arbitrary binning of data, but a decrease (or increase) in the slope of the decoding accuracy across trials. Moreover, the decoding analyses seem to be at the edge of SNR, hence making any interpretation that depends on the absence of an effect in some bins yet more problematic and implausible.

Thank you for the helpful suggestion. As previously mentioned we fitted a logarithmic model to quantify the change of the decoding benefit over trials, then found the trial index for which the change of the logarithmic fit was < 0.1 %. Given the results of this analysis and to ensure a sufficient number of trials, we focussed our further analyses on bins 1-2 . This is explained in more detail in the revised manuscript.

Relatedly, based on the literature there is no reason to assume that the dampening effect disappears with more training, thereby placing more burden of proof on the present results. Indeed, key studies supporting the dampening account (including human fMRI and MEG studies, as well as electrophysiology in non-human primates) usually seem to entail more learning than has occurred in bin 2 of the present study. How do the authors reconcile the observation that more training in previous studies results in significant dampening, while here the dampening effect is claimed to disappear with less training?

The discussion of these findings has been expanded on in the revised manuscript. As previously outlined, many of the studies supporting dampening did not explicitly test the effect of learning as they emerge, nor did they control for RS to the same extent.

The Methods section is quite bare bones. This makes an exact replication difficult or even impossible. For example, the sections elaborating on the GLM and cluster-based FWE correction do not specify enough detail to replicate the procedure. Similarly, how exactly the time points for significant decoding effects were determined is unclear (e.g., p. 11). Relatedly, the explanation of the decoding analysis, e.g. the choice to perform PCA before

decoding, is not well explained in the present iteration of the manuscript. Additionally, it is not mentioned how many PCs the applied threshold on average resulted in.

Thank you for this suggestion, we have described our methods in more detail.

To me, it is unclear whether the PCA step, which to my knowledge is not the default procedure for most decoding analyses using EEG, is essential to obtain the present results. While PCA is certainly not unusual, to my knowledge decoding of EEG data is frequently performed on the sensor level as SVMs are usually capable of dealing with the (relatively low) dimensionality of EEG data. In isolation this decision may not be too concerning, however, in combination with other doubts concerning the methods and results, I would suggest the authors replicate their analyses using a conventional decoding approach on the sensory level as well.

Thank you for this suggestion, we have explained our decision to use PCA in the revised manuscript.

Several choices, like the binning and the focus on bins 1-2 seem rather post-hoc. Consequently, frequentist statistics may strictly speaking not be appropriate. This further compounds above mentioned concerns regarding the reliability of the results.

The reasoning behind our decision to focus on bins 1-2 is now explained in more detail in the revised manuscript.

A notable difference in the present study, compared to most studies cited in the introduction motivating the present experiment, is that categories instead of exemplars were predicted.

This seems like an important distinction to me, which surprisingly goes unaddressed in the Discussion section. This difference might be important, given that exemplar expectations allow for predictions across various feature levels (i.e., even at the pixel level), while category predictions only allow for rough (categorical) predictions.

The decision to use categorical predictions over exemplars lies in the issue of RS, as it is impossible to control for RS while repeating stimuli over many trials. This has been discussed in more detail in the revised manuscript.

While individually minor problems, I noticed multiple issues across several figures or associated figure texts. For example: Figure 1C only shows valid and invalid trials, but the figure text mentions the neutral condition. Why is the neutral condition not depicted but mentioned here? Additionally, the figure text lacks critical information, e.g. what the asterisk represents. The error shading in Figure 2 would benefit from transparency settings to not completely obscure the other time-courses. Increasing the figure content and font size within the figure (e.g. axis labels) would also help with legibility (e.g. consider compressing the time-course but therefore increasing the overall size of the figure). I would also recommend using more common methods to indicate statistical significance, such as a bar at the bottom of the time-course figure typically used for cluster permutation results instead of a box. Why is there no error shading in Figure 2A but all other panels? Fig 2C-F has the y-axis label "Decoding accuracy (%)" but certainly the y-axis, ranging roughly from 0.2 to 0.7, is not in %. The Figure 3 figure text gives no indication of what the error bars represent, making it impossible to interpret the depicted data. In general, I would recommend that the authors carefully revisit the figures and figure text to improve the quality and complete the information.

Thank you for the suggestions. Figure 1C now includes the neutral condition. Asterisks denote significant results. The font size in Figure 2C-E has been increased. The y-axis on Figure 2C-E has been amended to accurately reflect decoding accuracy in percentage. Figure 2A has error shading, however, the error is sufficiently small that the error shading is difficult to see. The error bars in Figure 3 have been clarified.

Given the choice of journal (eLife), which aims to support open science, I was surprised to find no indication of (planned) data or code sharing in the manuscript.

Plans for sharing code/data are now outlined in the revised manuscript.

While it is explained in sufficient detail later in the Methods section, it was not entirely clear to me, based on the method summary at the beginning of the Results section, whether categories or individual exemplars were predicted. The manuscript may benefit from clarifying this at the start of the Results section.

Thank you for this suggestion, following this and suggestions from other reviewers, the experimental paradigm and the mappings between categories has been further explained in the revised manuscript, to make it clearer that predictions are made at the categorical level.

"Unexpected trials resulted in a significantly increased neural response 150 ms after image onset" (p.9). I assume the authors mean the more pronounced negative deflection here. Interpreting this, especially within the Results section as "increased neural response" without additional justification may stretch the inferences we can make from ERP data; i.e. to my knowledge more pronounced ERPs could also reflect increased synchrony. That said, I do agree with the authors that it is likely to reflect increased sensory responses, it would just be useful to be more cautious in the inference.

Thank you for the interesting comment, this has been rephrased as a "more pronounced negative deflection" in the revised manuscript.

Why was the ERP analysis focused exclusively on Oz? Why not a cluster around Oz? For object images, we may expect a rather wide dipole.

Feuerriegel et al (2021) have outlined issues questioning the robustness of univariate analyses for ES, as such we opted for a targeted ROI approach on the channel showing peak amplitude of the visually evoked response (Fig. 2B). More details on this are in the revised manuscript.

How exactly did the authors perform FWE? The description in the Method section does not appear to provide sufficient detail to replicate the procedure.

FWE as implemented in SPM is a cluster-based method of correcting for multiple comparisons using random field theory. We have explained our thresholding methods in more detail in the revised manuscript.

If I misunderstand the authors and they did indeed perform standard cluster permutation analyses, then I believe the results of the timing of significant clusters cannot be so readily interpreted as done here (e.g. p.11-12); see: Maris & Oostenveld 2007; Sassenhagen & Dejan 2019.

All statistics were based on FWE under random field theory assumptions (as implemented in SPM) rather than on cluster permutation tests (as implemented in e.g. Fieldtrip)

Why did the authors choose not to perform spatiotemporal cluster permutation for the ERP results?

As mentioned above, we opted to target our ERP analyses on Oz due to controversies in the literature regarding univariate effects of ES (Feuerriegel et al., 2021).

Some results, e.g. on p.12 are reported as T29 instead of Tmax. Why?

As mentioned above, prediction decoding analyses have been removed from the manuscript.

<https://doi.org/10.7554/eLife.103689.2.sa0>